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Surpassing stock market's noise-nonstationarity tradeoff by causal-based domain discovery

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While neural networks have been extensively applied in stock prediction, forecasting stock returns remains a significant challenge due to the nonstationary nature of financial markets. Existing models struggle to balance two competing demands: learning from large historical datasets to reduce noise, and maintaining temporal relevance in ever-changing market regimes. To address these challenges, we propose a novel *Domain Discovery* framework that leverages causal relationships to identify training samples from historical market environments that closely resemble the target prediction period. This selection mechanism enables models to learn more effectively from nonstationary data by avoiding outdated or mismatched information, while significantly increasing the number of useful labeled samples. Further, to avoid non-implementable alphas (i.e. profits unachievable under real-world market constraints), embedded within the Domain Discovery framework is a quantile-focused regularization technique that prioritizes the top-performing predictions, aligning training with practical investment goals. The entire approach is model-agnostic and integrates with a wide range of neural network architectures. Extensive experiments across six state-of-the-art architectures demonstrate consistent improvements in out-of-sample performance, underscoring the robustness and versatility of the proposed framework.

Keywords: Stock prediction; Transfer learning; Domain adaptation; Causality

JEL Classifications: C45, G11

1. Introduction

Neural-network-based stock prediction has become a powerful tool for capturing the intricate dynamics of financial markets, enabling actionable insights to be extracted from vast time-series and cross-sectional data (Jiang 2021). Advanced architectures, including Long Short-Term Memory (LSTM) networks (Hochreiter and Schmidhuber 1997), Convolutional Neural Networks (CNNs) (He *et al.* 2016) and Graph Neural Networks (GNNs) (Veličković *et al.* 2017, Sawhney *et al.* 2021, Xu *et al.* 2021), have shown remarkable capabilities in modeling nonlinear relationships and capturing temporal and cross-sectional dependencies. However, training these models effectively requires large datasets to prevent overfitting and ensure stability, especially when handling high-dimensional financial data. This challenge is further compounded by the inherent nonstationarity of financial markets, driven by factors such as economic shocks,

regulatory changes, and evolving investor sentiment (Baldovin *et al.* 2011).

For instance, figure 1(a) demonstrates that when training on datasets with an excessively long historical horizon, represented by LongHOR in figure 1(b), the Transformer architecture (Vaswani 2017) yields no profit. To address this, many studies adopt rolling updates, as illustrated in figure 1(c), which iteratively update models using the latest data (Kim *et al.* 2019, Nguyen and Yoon 2019), mitigating nonstationarity and narrowing the gap between in-sample and out-of-sample (OOS) predictions. Under this approach, the Transformer architecture begins to generate profits, as shown in figure 1(a). While rolling updates adapt to rapidly changing market conditions, they also increase sensitivity to noise due to smaller sample sizes. This introduces a critical trade-off: larger datasets improve robustness to noise and sample diversity but risk incorporating outdated information, whereas smaller, more recent datasets capture relevant patterns but are prone to overfitting.

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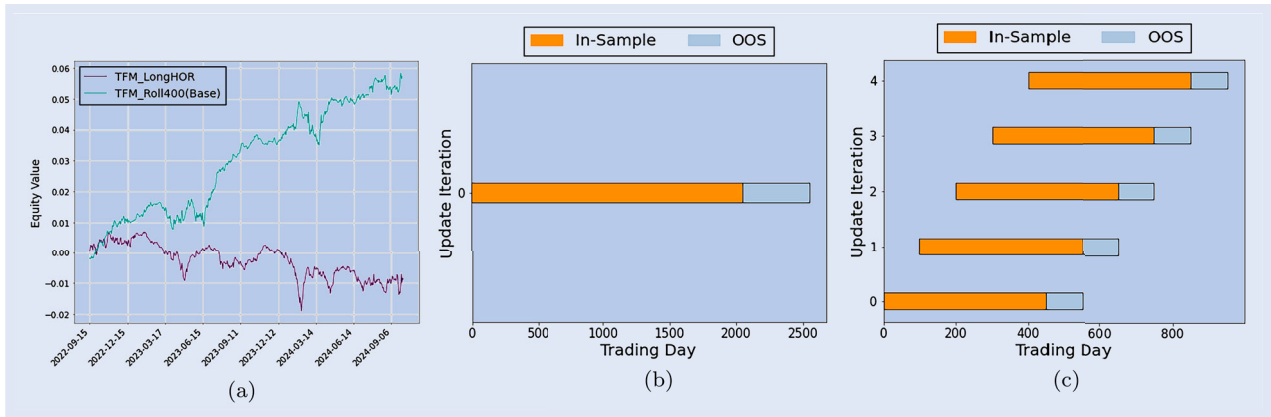


Figure 1. (a) Equity curves generated by the LongHOR and Roll400 schemes for the Transformer architecture, with models trained using an MSE-based learning objective. (b) The model is trained once using 2050 historical trading days and applied to a 500-trading-day OOS horizon. (c) The model is re-trained every 100 OOS trading days using the most recent 450 in-sample days, with the validation set comprising the last 50 days of the in-sample horizon, resulting in 400 training days for each update. (a) LongHOR vs. Roll400 for Transformer Encoder. (b) LongHOR and (c) Roll400.

However, rolling-window methods struggle to optimally balance these factors, as neither the degree of noise nor the timing of structural market shifts can be reliably estimated. Even with an optimally chosen window, the resulting model often represents a compromised solution rather than a robust one. To improve robustness, it is natural to consider techniques from Transfer Learning, which aim to transfer knowledge from historical environments to the most recent market environment where limited observations are available. Nevertheless, classical Transfer Learning methods, such as Covariate Shift Adaptation (Sugiyama *et al.* 2007) and Domain Adaptation (DA) (Zhao *et al.* 2019), are ineffective for stock prediction problems. For instance, Covariate Shift Adaptation assumes that the input distribution $\mathbb{P}(X)$ varies across environments while the conditional distribution $\mathbb{P}(Y|X)$, i.e. the causal mechanism, remains invariant. This assumption does not hold under market nonstationarity, where even the causal mechanisms evolve over time. Similarly, most DA methods implicitly rely on the near-invariance of $\mathbb{P}(Y|X)$, making them unsuitable for our problems. These challenges motivate the need to identify historical environments that most closely resemble the recent market so that relevant knowledge can be effectively transferred. This idea appears superficially related to earlier finance-specific regime-discovery approaches, such as change-point detection methods that segment markets into contiguous regimes (Matteson and James 2014), correlation-structure clustering of market states (Liao 2005), and volatility-regime models (Calvet and Fisher 2004). However, these approaches focus on macro-level regime segmentation for descriptive analysis or volatility forecasting, rather than identifying environments suitable to model learning. In contrast, the suitability of environments for model learning is determined by micro-structural features and causal dependencies induced by the stock prediction model itself. To address this gap, we propose the *Domain Discovery* framework, which maximizes the utility of historical data while mitigating the adverse effects of market nonstationarity. The method first partitions historical data into candidate training domains and then selects those most relevant to the out-of-sample (OOS) environment based on

causal relationships and distributional similarity tailored to the specific choice of prediction models. Unlike fixed rolling windows, which either incorporate outdated patterns or overfit to short-term noise, our framework ensures that model learning occurs only within selected domains that most closely resemble the OOS environment, thereby expanding effective training data while maintaining robustness to both noise and regime shifts.

Furthermore, this work places particular emphasis on the profitability of the top quantiles of model outputs. Instead of seeking models that perform uniformly well across all stocks, we prioritize those that excel in predicting large outcomes. That aligns with the practical objective of focusing on a subset of the most profitable stocks rather than diversifying across the entire stock universe. Table 1 presents a four-stock toy example comparing two prediction models evaluated using mean squared error (MSE). While both models achieve the same MSE, their predictions differ significantly in utility: investing in the top 50% of stocks based on Model 2 predictions yields a profit of 15%, compared to only 6% from equally investing across all stocks as suggested by Model 1 predictions. Beyond the benefits demonstrated in table 1, adopting a top-quantile approach offers additional advantages. First, reducing the number of stock holdings lowers the minimum capital requirements. For example, in the Chinese A-share market, stocks are traded in lots, with each lot typically costing over 2,000 US dollars. More importantly, holding long positions in top-quantile stocks is more likely to yield implementable alphas (i.e. profits achievable under real-world market constraints) compared to holding stocks in other quantiles. Many alphas, particularly those derived by shorting the bottom-quantile stocks, are non-implementable due to high costs or regulatory restrictions, rendering them practically irrelevant despite their theoretical existence. The high approximation ability of machine learning models can inadvertently bias predictions toward these non-implementable alphas, reducing their practical utility. Despite the significance of capturing top-quantile stocks, this aspect is often overlooked in academic research. Fortunately, our Domain Discovery framework inherently addresses this issue by enabling

Table 1. A four-stock toy example. The MSE for both model predictions is 0.010625.

Stock	True Return	Model 1 Prediction	Model 2 Prediction
A	0.1	0.1	0.2
B	-0.05	0.1	0.05
C	0.2	0.1	0.2
D	0	0.1	0.15

prediction models to prioritize the profitability of the top quantiles. Specifically, we regularize the model by aligning the marginal distributions of stock features with a customized prior distribution tailored to capture profits from the desired stock quantiles.

In summary, this work makes three key contributions:

- We propose a Domain Discovery method, grounded in rigorous theoretical analysis, that substantially increases the number of labeled samples while preserving the model's ability to learn from nonstationary market conditions.
- We introduce a quantile-specific method that is naturally integrated into the proposed Domain Discovery framework.
- We demonstrate the plug-and-play nature of our approach by applying it to six diverse state-of-the-art (SOTA) neural network architectures. Numerical results consistently show that our method enhances performance regardless of the underlying neural network architecture.

2. Analysis framework

Econometric studies suggest that the returns of stock n , denoted by r_n , can be decomposed into a causal-effect relationship involving the causal features (i.e. stock characteristics), $\mathbf{z}_n := (z_n^1, \dots, z_n^K)^\top$, the causal coefficients (i.e. factor premiums), $\mathbf{f} := (f^1, \dots, f^K)^\top$, and a residual term $\epsilon_n \in \mathbb{R}$ (Chincarini 2006):

$$r_n = z_n^1 f^1 + \dots + z_n^K f^K + \epsilon_n. \quad (1)$$

Our analytical framework is based on (1), where the prediction models aim to capture the $\mathbf{z}_n^\top \mathbf{f}$ component of stock returns, with our Domain Discovery method leveraging the causal perspective inherent in (1). This study focuses on the single-factor case ($K = 1$), though it can be easily extended to the multi-factor case by training multiple models using diverse architectures and performing post-training operations, such as ensembling or concatenating their outputs.

2.1. Settings & prediction error bound

Consider time steps $\{t_{-T-1}, t_{-T}, \dots, t_{-1}, t_0\}$, where $t_{-T-1} < t_{-T} < t_{-T+1} < \dots < t_{-1} < t_0$ for $T > 0$. The market information within each interval $(t_{-i-1}, t_{-i}]$ defines a market environment, indexed by t_{-i} . Specifically, each t_{-i} is treated as

a distinct domain and consists of $N_{t_{-i}} > 0$ input-label pairs, denoted by $\{(x_{nt_{-i}}, r_{nt_{-i}})\}_{n=1}^{N_{t_{-i}}}$, where $r_{nt_{-i}}$ represents the 1-day forward return of sample n in domain t_{-i} . We focus on domain-wise demeaned returns, which represent excess returns relative to the domain average. We further standardize these demeaned returns by dividing them by the standard deviation of returns over the entire in-sample horizon, as we empirically found that this normalization facilitates an easier hyperparameter tuning process. Here, $N_{t_{-i}}$ is defined as the product of the cardinality of the stock pool and the number of days within the interval $(t_{-i-1}, t_{-i}]$. The interval $(t_{-1}, t_0]$ represents the future horizon where the prediction model is applied, and t_0 is thus referred to as the *Out-Of-Sample (OOS) domain*. The domains t_{-i} for $i \neq 0$ are the candidate in-sample domains used for model training. Since each in-sample domain t_{-i} for $i \neq 0$ typically spans only a few days (e.g. 4 days in our experiments), we refer to it as a *micro-domain*. In contrast, the OOS domain t_0 often spans over more days, depending on the model update frequency (e.g. 100 days in our experiments).

This work builds upon the causal framework for the stock market proposed by Xu *et al.* (2025); we begin by briefly introducing it through the Structural Causal Model (SCM):

SCM of Stock Market. We adapt the factor model of the form (1), which characterizes the cause-effect relationship between stock returns and characteristics, into the form of a Structural Causal Model (SCM). Specifically, for a given domain t , the stock return after demeaning can be expressed as follows:

$$\begin{aligned} R_t &= \tilde{\mathbf{Z}}_t \tilde{\mathbf{F}}_t + \tilde{\epsilon}_t, \\ X_t^a &\leftarrow R_t, \\ X_t^a &\dashrightarrow X_t \leftarrow \tilde{\mathbf{Z}}_t, \end{aligned} \quad (2)$$

where

$$\tilde{\mathbf{Z}}_t, \tilde{\mathbf{F}}_t, \tilde{\epsilon}_t \sim \mathbb{P}_t.$$

Here, the random variable $\tilde{\mathbf{Z}}_t$ takes values in $\mathbb{R}$, representing the one-dimensional *causal feature*. It satisfies $\mathbb{E}_t[\tilde{\mathbf{Z}}] = 0$ and $\mathbb{E}_t[\tilde{\mathbf{Z}}^2] = 1$, where $\mathbb{E}_t[\tilde{\mathbf{Z}}] := \mathbb{E}_{\tilde{\mathbf{Z}} \sim \mathbb{P}_t}[\tilde{\mathbf{Z}}]$ for simplicity. $\tilde{\mathbf{F}}_t \in \mathbb{R}$ denotes the *causal coefficients*, which are often treated as deterministic within each t , reflecting the significance of each causal feature. The one-dimensional random variable $\tilde{\epsilon}_t$ is the residual term, satisfying $\tilde{\epsilon}_t \perp \tilde{\mathbf{Z}}_t$ and $\mathbb{E}_t[\tilde{\epsilon}] = 0$. X_t^a represents the *spurious feature*. The dashed arrows signify the variability of the relationship between X_t^a and R_t across environments. The observable stock information X_t encompasses both causal and spurious features. The random variables are jointly sampled according to the micro-domain measure $\mathbb{P}_t$ where the subscript t emphasizes the domain-variant nature of these random variables. To avoid ambiguity, uppercase letters denote random variables, while the corresponding lowercase letters are used for dummy variables in the context of functions and probability measures (e.g. $\mathbb{P}_t(\tilde{\mathbf{z}})$ and $g(x)$).

Let Ω^g denote the set of prediction models, defined as functions of the form $g(x) = \phi(x)\mathbf{v}$, where $\phi(x)$ and $\mathbf{v}$ take values in $\mathbb{R}$. The primary objective is to minimize the error on the OOS domain t_0 , expressed as $\mathbf{err}_{t_0}^g := \mathbb{E}_{t_0}[\ell(g(x), r)]$, where $\ell: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}_+$ is a predefined distance function. In this work, we focus on the ℓ_1 -norm. Our method specifically aims

to reduce the following error bound (Xu *et al.* 2025):

$$\mathbf{err}_{t_0}(g) \leq \frac{1}{T} \sum_{i=1}^T \mathbf{err}_{t-i}^g + \frac{1}{T} \sum_{i=1}^T \mathcal{W}^1(\mathbb{P}_{t_0}(g(x), r), \mathbb{P}_{t-i}(g(x), r)) + 2\lambda^*, \quad (3)$$

where the upper bound consists of three components: the average training error, a Wasserstein distance term quantifying the distance between joint distributions, and the ideal joint error λ^* . The Wasserstein distance term is defined as:

$$\mathcal{W}^n(\mathbb{P}_{t_0}(x), \mathbb{P}_{t-i}(x)) := \inf_{\pi \in \Pi_{t-i}} \left(\int \|x_{t_0} - x_{t-i}\|_n^n d\pi \right)^{\frac{1}{n}}, \quad (4)$$

where Π_{t-i} represents the set of couplings between $\mathbb{P}_{t-i}(x)$ and $\mathbb{P}_{t_0}(x)$ (Villani 2009). The ideal joint error λ^* is defined as:

$$\lambda^* := \min_{g \in \Omega^g} \frac{1}{T} \sum_{i=1}^T \mathbf{err}_{t-i}^g + \mathbf{err}_{t_0}^g. \quad (5)$$

2.2. Relation to domain adaptation

The formulation of the generalization error bound (3) is motivated by the theoretical developments in the domain adaptation literature (Ben-David *et al.* 2010, Johansson *et al.* 2019). In particular, several studies have adopted Wasserstein distances to quantify the discrepancy between domains (Flamary *et al.* 2016, Shen *et al.* 2018). A common approach to reducing the OOS error is to learn feature representations that minimize such domain discrepancies (i.e. to minimize the Wasserstein term in (3)) as in Shen *et al.* (2018). However, counterexamples have shown that the OOS error can remain large even after applying these adaptation techniques (Zhao *et al.* 2019, Mahajan *et al.* 2021). This issue arises because such methods implicitly assume that λ^* remains small during the adaptation process, which is not generally true. Some works (Zhao *et al.* 2019, Arjovsky 2020, Sicilia *et al.* 2023) have identified the influence of λ^* , arguing that when the learned feature representations yield small domain discrepancies but λ remains large, it signals the occurrence of negative transfer. In particular, Zhao *et al.* (2019) derived a lower bound on the OOS error, showing that minimizing only the first two terms of (3) can actually increase the OOS error if the marginal label distributions differ significantly between the source and OOS domains.

In financial context, as demonstrated by Xu *et al.* (2025), λ^* also reflects the suitability of the training set for the learning task. Specifically, if the training set is heavily affected by market nonstationarity, λ^* will be large, and vice versa. While the other terms in the error bound can be effectively reduced by properly identifying causal features, the size of λ^* remains fixed once the training set is determined. A high exposure to market nonstationarity ensures a persistently large error bound, regardless of the learning techniques applied subsequently. Thus, achieving a smaller λ^* is a critical prerequisite for obtaining a lower error bound.

In addition, conventional Domain Adaptation methods are not applicable for reducing the Wasserstein term in (3) within

the stock prediction setting. Typically, such methods require access to (at least) the input marginal distributions of the out-of-sample (OOS) domain in order to align feature representations. However, in our temporal forecasting context, the marginal distribution of the OOS domain is unavailable until after the time point t_0 . Therefore, we address the problem under a more realistic scenario where the OOS domain is entirely unknown. Instead of relying on explicit access to the OOS distribution, we reduce the Wasserstein term from a Causal Perspective by identifying and training on in-sample domains whose causal mechanisms are most likely to remain invariant beyond t_0 .

2.3. Causal discovery

The Wasserstein term in (3) vanishes only when the SCM dynamics between the features induced by ϕ and the labels in the in-sample domains are identical to those in the OOS domain t_0 . This can be achieved by finding ϕ such that both $\mathbb{E}_t[R | \phi(X)]$ and $\mathbb{P}_t(\phi(x))$ remain invariant across t as suggested by the linear relationship between $\tilde{\mathbf{Z}}_t$ and R_t in (2). Under the linearity assumption, the invariance of $\mathbb{E}_t[R | \phi(X)]$ is equivalent to stating that the regression coefficient, denoted by:

$$\mathbf{v}_t^\phi := \frac{\mathbb{E}_t[\phi(X)R]}{\mathbb{E}_t[\phi(X)^2]}, \quad (6)$$

is invariant across t .[†] Notably, when $\phi(X_t) = \tilde{\mathbf{Z}}_t$, it follows that $\tilde{\mathbf{F}}_t = \mathbf{v}_t^\phi$, and thus $\mathbf{v}_t^\phi$ is referred to as the estimated causal coefficient. However, since the OOS domain t_0 is inaccessible, it is a common practice to discover invariance for $\mathbf{v}_t^\phi$ and $\mathbb{P}_t(\phi(x))$ across in-sample domains, with the assumption that such invariance is likely to persist into the OOS domain. This assumption is rooted in the idea that if a relationship consistently holds across a variety of different environments, this relationship represents the true causal link. This work adopts the perspective of Arjovsky *et al.* (2019), which asserts that causal discovery is equivalent to identifying invariance. Specifically, *Causal Discovery* in this context refers to learning ϕ that ensures invariant $\mathbf{v}_t^\phi$ and $\mathbb{P}_t(\phi(x))$ across in-sample micro-domains.

Furthermore, Causal Discovery based on identifying invariant $\mathbf{v}_t^\phi$ is inherently well-suited for stock prediction problems, as the assumption of an *Alpha-Generating Factor* closely aligns with our approach. As illustrated in figure 2, our causal feature $\tilde{\mathbf{Z}}_t$ is often referred to as an Alpha-Generating Factor in the financial industry when its corresponding causal coefficient $\tilde{\mathbf{F}}_t$, also known as the *Factor Premium*, is stable and positive for the majority of the time. Hedge funds aim to identify Alpha-Generating Factors, operating under the premise that if a factor's excess premium is both positive and stable over a historical horizon, its profitability is likely to persist in the future. Consequently, we adopt the financial industry's convention by focusing on the SCM (2), which incorporates causal features with predominantly positive causal coefficients (i.e. $\tilde{\mathbf{F}}_t > 0$ for most t). On the other

[†] Note that $\mathbf{v}_t^\phi$ minimizes the regression problem $\min_{\hat{\mathbf{v}}} \mathbb{E}_t[(R - \phi(x)\hat{\mathbf{v}})^2]$.

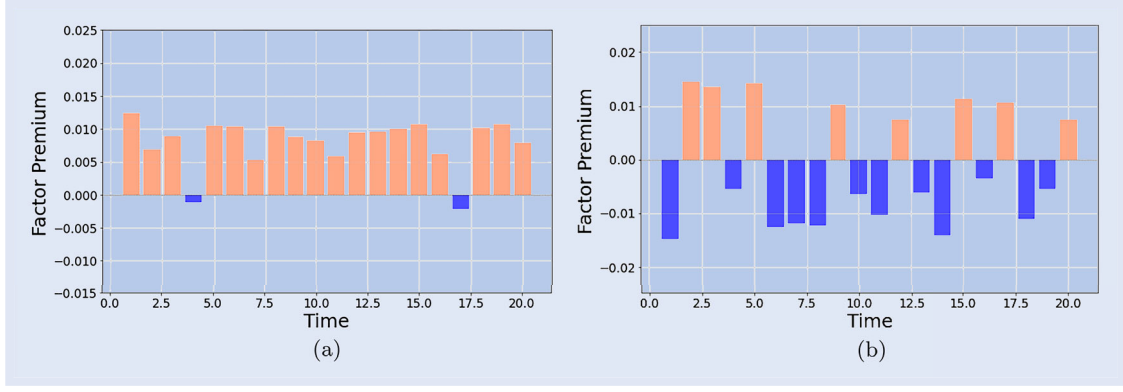


Figure 2. Conceptual comparison of Alpha-Generating and Non-Alpha-Generating Factors based on artificial data. Panel (a) shows an alpha-generating factor, which exhibits a relatively consistent and distinguishable pattern of factor premiums, suggesting potential predictive power. In contrast, Panel (b) depicts a non-alpha-generating factor, whose factor premiums fluctuate in a more random and noisy manner, lacking any discernible trend or persistent advantage. (a) Alpha-Generating Factor and (b) Non-Alpha-Generating Factor.

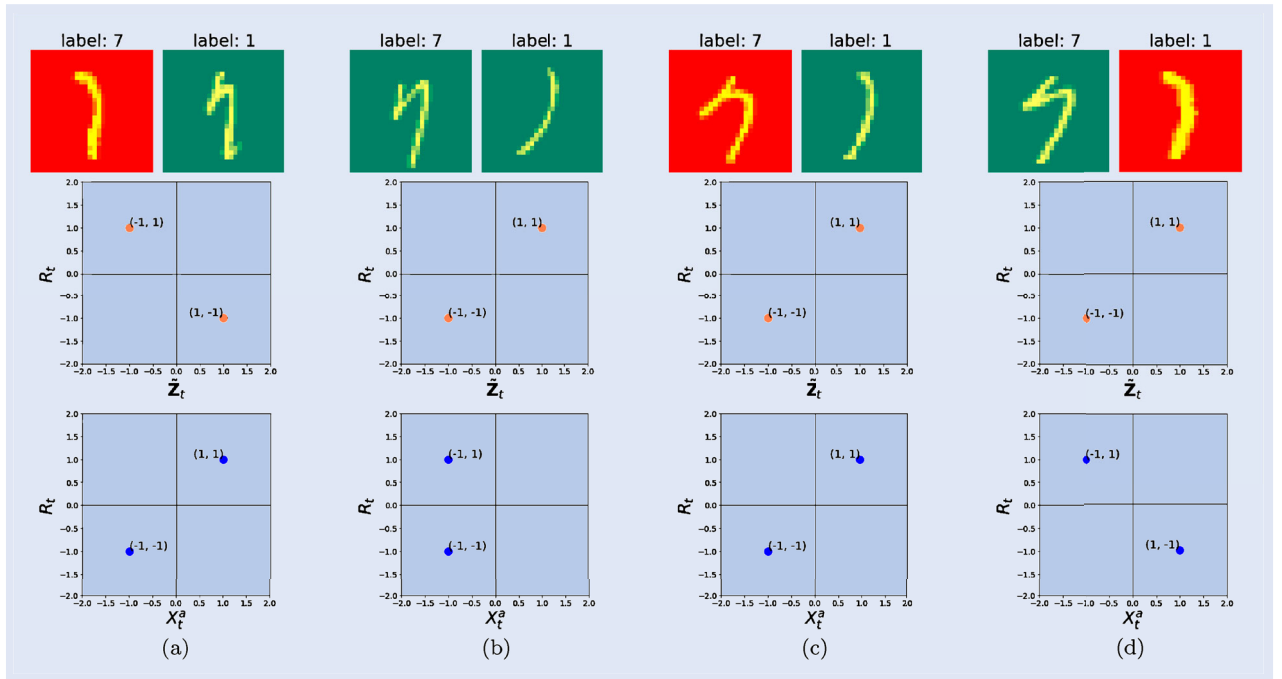


Figure 3. *Toy Example*: Three in-sample micro-domains t_{-3} , t_{-2} , t_{-1} , and one OOS micro-domain t_0 are considered, with each domain consisting of two images from the MNIST dataset. The shape of the digit in each image corresponds to the causal feature $\tilde{Z}_t$, while the background color represents the spurious feature X_t^a . Specifically, $\tilde{Z}_t = 1$ if the digit's shape involves a sharp bend, and $\tilde{Z}_t = -1$ otherwise. Similarly, $X_t^a = 1$ if the background is red, and $X_t^a = -1$ otherwise. The label R_t is defined as 1 if the true digit is '7', and $R_t = -1$ otherwise. The causal coefficients are $\tilde{F}_t = 1, 1, -1, 1$ for the domains t_{-3}, t_{-2}, t_{-1} , and t_0 , respectively. The images corresponding to $(\tilde{Z}_t, R_t)$ and (X_t^a, R_t) for each micro-domain are visualized in coordinate plots. (a) micro-domain t_{-3} : $\tilde{F}_t = -1$. (b) micro-domain t_{-2} : $\tilde{F}_t = 1$. (c) micro-domain t_{-1} : $\tilde{F}_t = 1$ and (d) OOS t_0 : $\tilde{F}_t = 1$.

hand, while invariant $\mathbb{P}_t(\phi(x))$ may not have a direct counterpart in industry practices, we demonstrate in section 3.3 that it plays a crucial role in guiding our prediction model to prioritize top-quantile performance.

2.4. MNIST example

To illustrate, we construct a toy example shown in figure 3, featuring three in-sample micro-domains t_{-3} , t_{-2} , and t_{-1} , along with one out-of-sample (OOS) micro-domain t_0 , representing four different individuals' handwriting. Each micro-domain contains two images selected from the MNIST dataset

(LeCun *et al.* 1998).[†] To simulate noise, we manually color the backgrounds of all images. In this setup, each image represents the model input X_t , the shape of the digit in the image corresponds to the causal feature $\tilde{Z}_t$, and the background color represents the spurious feature X_t^a . Specifically, the causal feature $\tilde{Z}_t$ takes the value 1 if the digit's shape exhibits a sharp bend and -1 otherwise. Similarly, the spurious feature X_t^a

[†] '7' and '1' from t_1 : the 436th and 39,354th images from the training set, respectively; '7' and '1' from t_2 : the 1689th image from the testing set and the 1080th image from the training set, respectively; '7' and '1' from t_3 : the 1080th and 4201th images from the testing set, respectively; '7' and '1' from t_0 : the 5280th and 58,022th images from the training set, respectively.

is assigned a value of 1 if the background color is red and -1 for any other color. The label R_t is defined as 1 if the true digit is '7', and -1 otherwise. For each micro-domain, the images are visualized in coordinate plots based on their respective feature-label mappings $(\tilde{\mathbf{Z}}_t, R_t)$ and (X_t^a, R_t) , effectively demonstrating the relationships between the causal and spurious features across domains. Furthermore, suppose our prediction model is of the form $g(x) = \phi(x)\mathbf{v}$, where $\phi(x) \in \{1, -1\}$ and $\mathbf{v} \in \mathbb{R}$. Let us consider MSE as both the learning objective function and the OOS evaluation metric.

Influence from λ^ .* Notably, the micro-domain t_{-3} differs significantly from the others in terms of $\tilde{\mathbf{Z}}_t$, where the digit '1' resembles the digit '7' from the other domains, and vice versa. If the training set includes all three micro-domains t_{-1} , t_{-2} , and t_{-3} , the ideal joint error can be verified as $\lambda^* = 0.667$. Furthermore, using $\phi(X_t) = X_t^a$ leads to a lower MSE on the training set (i.e. $MSE \approx 1.111$ with $\mathbf{v} = \frac{2}{3}$) compared to $\phi(X_t) = \tilde{\mathbf{Z}}_t$ (i.e. $MSE \approx 1.778$ with $\mathbf{v} = \frac{1}{3}$). Consequently, training on all micro-domains favors the spurious feature, X_t^a , over the causal feature, $\tilde{\mathbf{Z}}_t$, resulting in the background color being used to predict the label. This approach, however, fails completely on the OOS domain t_0 . On the other hand, if only t_{-1} and t_{-2} are included in the training set, the ideal joint error λ^* vanishes, and the training MSE for $\phi(X_t) = \tilde{\mathbf{Z}}_t$ with $\mathbf{v} = 1$ reduces to zero, whereas the training MSE for $\phi(X_t) = X_t^a$ does not. As a result, the prediction model leverages the causal features on t_0 , achieving a zero OOS MSE. Indeed, it may be uncommon for someone's handwriting to be as unusual as that displayed in t_{-3} , but shifts in the causal relationship between factors and stock returns occur far more frequently in real markets due to market nonstationarity. For example, the momentum factor, which drives profits during a bull market, can become a source of losses in a bear market.

Domain Adaptation. Since the OOS domain t_0 is unavailable in stock prediction tasks, we simulate the domain adaptation process by transferring knowledge from historical domains t_{-3} and t_{-2} to t_{-1} . Specifically, we approximate the Wasserstein term in (3) by replacing t_0 with t_{-1} and learn the feature representations that minimize this term, while setting $\mathbf{v} = \tilde{\mathbf{F}}_{t_{-1}} = 1$. In this setting, the mapping $\phi(X_t) = X_t^a$ minimizes the Wasserstein term because $\mathcal{W}^1(\mathbb{P}_{t_{-1}}(X^a, r), \mathbb{P}_{t_{-2}}(X^a, r))$ only needs to transport the point $(-1, 1)$ to $(1, 1)$, whereas $\mathcal{W}^1(\mathbb{P}_{t_{-1}}(\tilde{\mathbf{Z}}, r), \mathbb{P}_{t_{-2}}(\tilde{\mathbf{Z}}, r))$ must also transport an additional point $(1, -1)$ to $(-1, -1)$. Consequently, applying domain adaptation here causes ϕ to prioritize minimizing the Wasserstein distance by exploiting spurious correlations, thereby learning spurious rather than causal features.

Causal Discovery. If t_{-1} and t_{-2} are chosen as the in-sample micro-domains in the toy example (figure 3), the Wasserstein term in (3) vanishes when $\phi(X_t) = \tilde{\mathbf{Z}}_t$, as $\mathbb{P}_t(\tilde{\mathbf{z}}, r)$ remains invariant for $t = t_{-1}, t_{-2}$, or t_0 . Concurrently, the corresponding $\mathbf{v}_t^\phi \equiv 1$, and $\mathbb{P}_t(\phi(x))$ is also invariant. In contrast, if $\phi(X_t) = X_t^a$, then $\mathbf{v}_{t_{-1}}^\phi = 1$ and $\mathbf{v}_{t_{-2}}^\phi = 0$. Therefore, Causal Discovery based on identifying in-sample invariance effectively filters out the spurious features X_t^a , resulting in a vanishing error bound. Conversely, Causal Discovery fails when the micro-domain t_{-3} is included, as the ideal joint error λ^* becomes non-zero. Specifically, if the in-sample domains

are t_{-1} and t_{-3} , X_t^a will be selected as the discovered feature since the corresponding ϕ results in $\mathbf{v}_{t_{-1}}^\phi = \mathbf{v}_{t_{-3}}^\phi$. Analogously, minimizing the in-sample domain's exposure to nonstationarity is a prerequisite for validating Causal Discovery in real markets.

3. Methodology with theoretical analysis

3.1. Macro environment splitting

In this section, we present a methodology designed to minimize the ideal joint error while ensuring an adequate sample size. Firstly, we divide the entire in-sample horizon into two sub-regions, as illustrated in figure 4. Specifically, let $\mathcal{T}_{trg} := \{t_{-i}\}_{0 < i \leq T}$ and $\mathcal{T}_{hist} := \{t_{-i}\}_{T < i \leq T'}$, where $0 < T < T'$. Then, the first region $\mathcal{T}_{trg}$ comprises the micro-domains closest to the OOS domain t_0 , referred to as the *target domain*, while the second region $\mathcal{T}_{hist}$ includes the remaining historical micro-domains. The target domain typically spans far fewer trading days compared to the second region (e.g. 100 days versus 2,400 days in our experiments). Based on prior knowledge of market nonstationarity, the target domain's environment is presumed to be more similar to that of the OOS domain. To capitalize on this, we employ a *Domain Discovery Method* to select micro-domains from the second region to form the *source domain*, ensuring its environment closely resembles that of the target domain. This approach enables the model to leverage samples from the target domain while being further augmented with those from the source domain, achieving both a smaller ideal joint error and a larger sample size.

Specifically, we formalize source and target domains using domain measures. For $t \in \mathcal{T}_{trg}$, we assume that the micro-domain distribution $\mathbb{P}_t$ is uniformly sampled from the target distribution, denoted by $\hat{\mathbb{P}}_{trg}$, with probability $1/|\mathcal{T}_{trg}|$. Similarly, let $\boldsymbol{\alpha} := (\alpha_{t_{-T-1}} \cdots \alpha_{t_{-T'}})^\top \in \{0, 1\}^{|\mathcal{T}_{hist}|}$, where each binary α_t represents the selection of a micro-domain. For $t \in \mathcal{T}_{hist}$, the distribution $\mathbb{P}_t$ is sampled from $\hat{\mathbb{P}}_{src}^\alpha$ with probability $\frac{\alpha_t}{\sum_{t' \in \mathcal{T}_{hist}} \alpha_{t'}}$. Thus, $\boldsymbol{\alpha}$ determines the selection of micro-domains, and $\hat{\mathbb{P}}_{src}^\alpha$ depends on $\boldsymbol{\alpha}$. This formulation implies that micro-domains are generated by a macro-environment, and we extend the in-sample SCM (2) by appending an additional statement: $\mathbb{P}_t \sim \hat{\mathbb{P}}_e$ for $e \in \{src, trg\}$, omitting the superscript $\boldsymbol{\alpha}$ for simplicity. The prediction error with respect to e is then expressed as:

$$\mathbf{err}_e(g) := \mathbb{E}_{X, R \sim \hat{\mathbb{P}}_e} [l(g(X), R)] = \int \int l(g(x), r) d\mathbb{P}_t d\hat{\mathbb{P}}_e, \quad \forall e \in \{src, trg\}, \quad (7)$$

and the ideal joint error is defined as:

$$\lambda_e^* := \min_{g \in \Omega_g} \mathbf{err}_e(g) + \mathbf{err}_{t_0}(g), \quad \forall e \in \{src, trg\}. \quad (8)$$

Our methodology is based on the assumption that the target domain closely resembles the OOS domain. Formally:

ASSUMPTION 3.1 The population minimizer (i.e. optimized with infinite sample size) on the target domain is still an

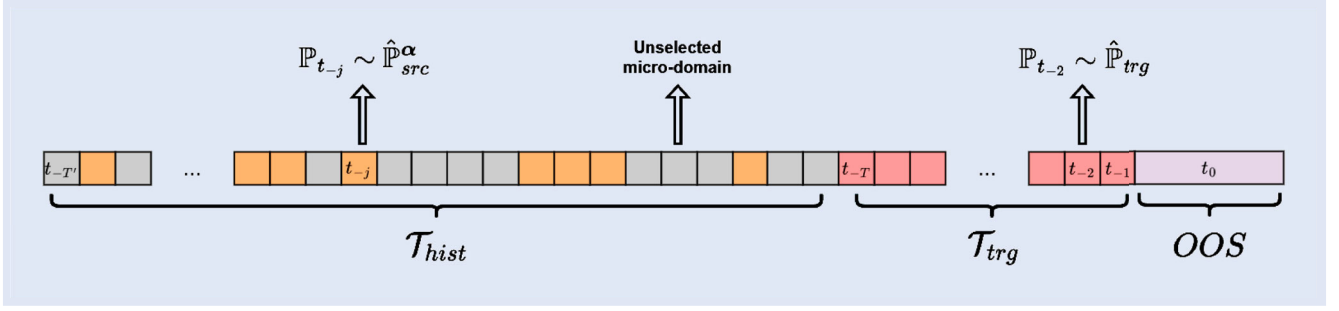


Figure 4. Micro-domains in $\mathcal{T}_{trg}$, used for training, are highlighted in pink. Selected micro-domains from $\mathcal{T}_{hist}$ that contribute to training are shown in orange, while unselected micro-domains are depicted in grey.

effective minimizer when the OOS domain is also included:

$$\arg \min_{g \in \Omega^s} \mathbf{err}_{trg}(g) \subset \arg \min_{g \in \Omega^s} \mathbf{err}_{trg}(g) + \mathbf{err}_{t_0}(g). \quad (9)$$

3.2. Domain discovery

With the Wasserstein distance of $n = 1$ defined in (4), we present the error bounds when considering the macro environments:

PROPOSITION 3.1 *For an OOS domain t_0 , the OOS prediction error is bounded above as follows:*

$$\mathbf{err}_{t_0}(g) \leq \mathbf{err}_e(g) + \mathcal{W}^1(\hat{\mathbb{P}}_e(g(x), r), \mathbb{P}_{t_0}(g(x), r)) + 2\lambda_e^*, \quad \forall e \in \{src, trg\}. \quad (10)$$

Proof Appendix 1. ■

Also, it is straightforward to get Corollary 3.1.1 by averaging the source and target error bounds:

COROLLARY 1 *When considering both the source and target domains, the error bound is given by:*

$$\mathbf{err}_{t_0}(g) \leq \frac{1}{2} \sum_{e \in \{src, trg\}} \mathbf{err}_e(g) + \frac{1}{2} \sum_{e \in \{src, trg\}} \mathcal{W}^1(\hat{\mathbb{P}}_e(g(x), r), \mathbb{P}_{t_0}(g(x), r)) + \lambda_{src}^* + \lambda_{trg}^*. \quad (11)$$

The target-only case of bound in (10) can be interpreted as a macro-domain version of (3). By selecting a small training horizon close to t_0 , we can maintain a small λ_{trg}^* ; however, a limited sample size may result in poor performance of g . In contrast, the joint scenario (11) incorporates the source domain, providing additional data and mitigating the effects of noise. Next, we introduce the Domain Discovery method, which demonstrates that incorporating the source domain can still achieve minimal exposure to the risks of nonstationarity.

Domain Discovery refers to the process of identifying $\hat{\mathbb{P}}_{src}^\alpha$ such that the ideal joint error is not increased when incorporating samples from both the source and target domains, compared to training models exclusively on the target domain. To be specific, the ideal joint error in (11) should not exceed that of (10) for $e=trg$ (i.e. $\lambda_{src}^* + \lambda_{trg}^* \leq 2\lambda_{trg}^*$). We begin by introducing Proposition 3.2, which demonstrates that if there exists some α satisfying $\mathcal{W}^1(\hat{\mathbb{P}}_{src}^\alpha(g_{trg}^*(x), r), \hat{\mathbb{P}}_{trg}(g_{trg}^*(x), r)) = 0$, then the ideal joint error in (11) will be less than or equal

to that of the target-only scenario. This result implies that the impact of nonstationarity is not amplified, enabling effective utilization of additional samples from the source domain.

PROPOSITION 3.2 *Let $g_{trg}^* \in \arg \min_{g \in \Omega^s} \mathbf{err}_{trg}(g)$, then*

$$\lambda_{src}^* \leq \lambda_{trg}^* + \mathcal{W}^1(\hat{\mathbb{P}}_{src}^\alpha(g_{trg}^*(x), r), \hat{\mathbb{P}}_{trg}(g_{trg}^*(x), r)). \quad (12)$$

Proof Appendix 2. ■

The Domain Discovery task can therefore be formulated as the follow to match selected source and target micro-domains (figure 5):

$$\begin{aligned} \min_{\alpha \in \{0,1\}^{|\mathcal{T}_{hist}|}} \quad & \mathcal{W}^1(\hat{\mathbb{P}}_{src}^\alpha(g_{trg}^*(x), r), \hat{\mathbb{P}}_{trg}(g_{trg}^*(x), r)) \\ \text{subject to} \quad & \sum_{t \in \mathcal{T}_{hist}} \alpha_t = T_\alpha, \end{aligned} \quad (13)$$

where $T_\alpha > 0$ is a hyperparameter controlling the number of micro-domains within the discovered source domain.

Efficient Alternative. Despite the succinctness of (13), evaluating the Wasserstein distance between the joint probabilities is computationally expensive. Specifically, if each micro-domain contains N samples, even the Sinkhorn Algorithm (Sinkhorn 1967), a widely used efficient method for computing the Wasserstein distance, has a complexity of $\mathcal{O}(|\mathcal{T}_{hist}| |\mathcal{T}_{trg}| N^2 / \epsilon)$, where $\epsilon > 0$ represents the precision parameter such that $1/\epsilon$ determines the number of iterations in the algorithm. In our experiments, with $N = 1000$, direct computation becomes intractable due to the quadratic dependency on the sample size.

To address this challenge, we propose an *Efficient Alternative* to (13). Notably, the estimated causal coefficient can be treated as a random variable, i.e. $\mathbf{v}_t^\phi \sim \hat{\mathbb{P}}_e(\mathbf{v}_t^\phi)$, because $\mathbb{P}_t$ is a random variable with respect to $\hat{\mathbb{P}}_e$ for $e \in \{src, trg\}$. Then, with the Wasserstein distance of $n = 2$ defined in (4), the following result holds:

PROPOSITION 3.3 *Suppose that (i) $\phi(X_t)$ is Z-score standardized in each micro-domain t (i.e. $\mathbb{E}_t[\phi(X)] = 0$ & $\mathbb{E}_t[\phi(X)^2] = 1$), and (ii) $\mathbb{P}_t(\phi(x))$ is invariant $\forall t \in \mathcal{T}_{hist} \cup \mathcal{T}_{trg}$, then*

$$\begin{aligned} \mathcal{W}^1(\hat{\mathbb{P}}_{src}^\alpha(g_{trg}^*(x), r), \hat{\mathbb{P}}_{trg}(g_{trg}^*(x), r)) \\ \leq 2\mathcal{W}^2(\hat{\mathbb{P}}_{src}^\alpha(\mathbf{v}_t^\phi), \hat{\mathbb{P}}_{trg}(\mathbf{v}_t^\phi)) \end{aligned}$$

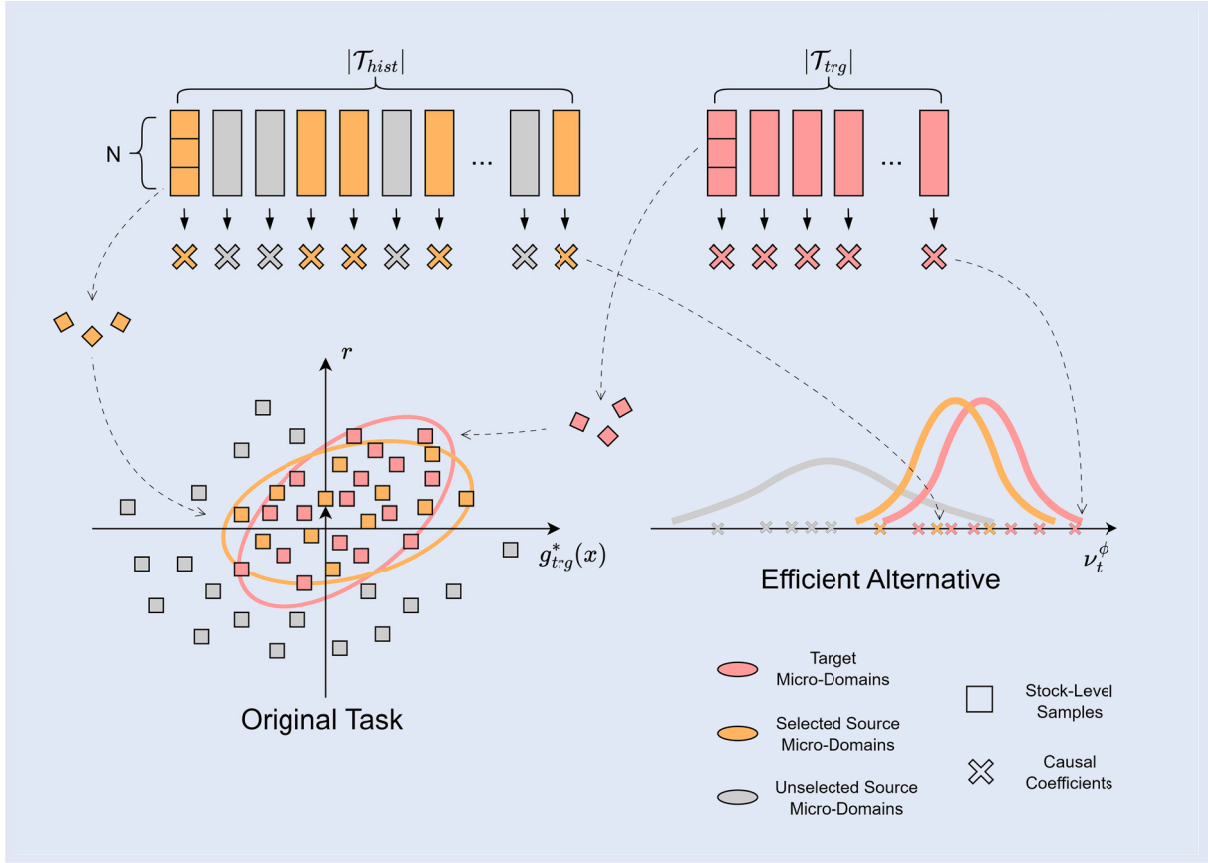


Figure 5. The *Domain Discovery* task is the process of selecting source micro-domains whose distributions most closely align with those of the target micro-domains. The *Original Task* (13) operates at the stock-level samples, involving $N \times (|\mathcal{T}_{hist}| + |\mathcal{T}_{trg}|)$ data points. The *Efficient Alternative* (15) performs a similar matching at the causal coefficient level, using only $(|\mathcal{T}_{hist}| + |\mathcal{T}_{trg}|)$ causal coefficients.

$$\begin{aligned}
 & + \sum_{e \in \{src, trg\}} \sqrt{\mathbb{E}_{\mathbf{v}_t^\phi \sim \hat{\mathbb{P}}_e} [(\mathbf{v}_{trg}^* - \mathbf{v}_t^\phi)^2]} \\
 & + \sum_{e \in \{src, trg\}} \sqrt{1 - \mathbb{E}_{\mathbf{v}_t^\phi \sim \hat{\mathbb{P}}_e} [(\mathbf{v}_t^\phi)^2]}, \quad (14)
 \end{aligned}$$

where $\mathbf{v}_{trg}^* := \mathbb{E}_{\mathbf{v}_t^\phi \sim \hat{\mathbb{P}}_{trg}} [\mathbf{v}_t^\phi]$.

Proof Appendix 3.

As a result, the *Domain Discovery Method* in this work is formulated by minimizing the upper bound of (14) while preserving conditions (i) and (ii) from Proposition 3.3:

$$\begin{aligned}
 & \min_{\alpha \in \{0,1\}^{|\mathcal{T}_{hist}|}} 2\mathcal{W}^2(\hat{\mathbb{P}}_{src}^\alpha(\mathbf{v}_t^\phi), \hat{\mathbb{P}}_{trg}(\mathbf{v}_t^\phi)) \\
 & + \sqrt{\mathbb{E}_{\mathbf{v}_t^\phi \sim \hat{\mathbb{P}}_{src}^\alpha} [(\mathbf{v}_{trg}^* - \mathbf{v}_t^\phi)^2]} + \sqrt{1 - \mathbb{E}_{\mathbf{v}_t^\phi \sim \hat{\mathbb{P}}_{src}^\alpha} [(\mathbf{v}_t^\phi)^2]} \\
 & \text{subject to } \sum_{t \in \mathcal{T}_{hist}} \alpha_t = T_\alpha. \quad (15)
 \end{aligned}$$

This formulation leads to an *Efficient Alternative* to the original Domain Discovery task (13). As shown in figure 5, unlike (13) where stock-level samples (i.e. $\{(g_{trg}^*(x_{nt}), r_{nt})\}_{n=1}^N$) are required for each micro-domain when computing the Wasserstein distance, the proposed approach in (15) relies solely on the estimated causal coefficient $\mathbf{v}^\phi$. Consequently, each micro-domain is associated with only

one sample of $\mathbf{v}^\phi$, reducing the computational complexity from $\mathcal{O}(|\mathcal{T}_{hist}||\mathcal{T}_{trg}|N^2/\epsilon)$ to $\mathcal{O}(|\mathcal{T}_{hist}||\mathcal{T}_{trg}|/\epsilon)$. In this work's setting, the Efficient Alternative is one million times more efficient compared with the original Domain Discovery task (i.e. $N^2 = 1,000,000$).

3.3. Invariance of $\mathbb{P}_t(\phi(x))$ and top-quantile performance

■ The significance of invariance of $\mathbb{P}_t(\phi(x))$ is twofold: (i) it is a fundamental component of Causal Discovery (section 2.3), and (ii) it is required for the Domain Discovery Method (15) (section 3.2). While the Z-score standardization of $\phi(X_t)$ —a prerequisite of our Domain Discovery Method—ensures that the first and second moments of $\phi(X_t)$ remain invariant through moment matching, achieving invariance for $\mathbb{P}_t(\phi(x))$ necessitates addressing higher-order moments beyond the second. In the Causal Discovery method proposed by Xu *et al.* (2025), the focus is limited to the first two moments, based on the argument that discrepancies from higher-order moments are negligible in comparison. Similarly, the financial industry often emphasizes the first four moments, with Z-score standardization of $\phi(X_t)$ being the most common preprocessing step. However, we observe that aligning all moments offers significant advantages. This approach not only provides stricter adherence to theoretical principles but also enhances top-quantile profits.

Notably, maximizing the correlation between $\phi(X_t)$ and returns is a key component of various learning objectives

in stock prediction problems. For instance, minimizing the mean-squared error between $g(X_t)$ and R_t is equivalent to minimizing $\mathbf{v}^2 - 2\mathbf{v}\text{Var}(R_t)^{\frac{1}{2}}\text{Corr}(\phi(X_t), R_t)$ for $g(x) = \phi(x)\mathbf{v}$; additionally, $\text{Corr}(\phi(X_t), R_t)$ itself serves as a learnable proxy for the information coefficient (Fang *et al.* 2020), which is a metric used to evaluate the effectiveness of a predictive signal in forecasting stock returns. Therefore, it suffices to conduct our analysis based on $\text{Corr}(\phi(X_t), R_t)$. Remark 3.1 suggests that such a correlation can be decomposed into terms associative with quantiles of $\phi(X_t)$.

REMARK 3.1 For $t \in \mathcal{T}_{\text{hist}} \cup \mathcal{T}_{\text{trg}}$ and $Q > 1$, let $\tau_1, \tau_2, \dots, \tau_{Q-1}$ be the quantiles of $\phi(X)$ such that $\mathbb{P}_t(\tau_{q-1} < \phi(X) < \tau_q) = \frac{1}{Q}$ where $q \in \{1, \dots, Q\}$ with $\tau_0 = -\infty$ and $\tau_Q = \infty$. Define $\Omega_t^{\phi, q} := \{\text{events s.t. } \tau_{q-1} < \phi(X_t) < \tau_q\}$. Then, $\text{Corr}(\phi(X_t), R_t)$ can be decomposed as the following:

$$\begin{aligned} & \text{Corr}(\phi(X_t), R_t) \\ &= \frac{1}{Q\text{Var}(R_t)^{\frac{1}{2}}} \sum_{q=1}^Q (\tilde{\mathbf{F}}_t \text{Var}(\phi(X_t) | \Omega_t^{\phi, q})^{\frac{1}{2}} \text{Var}(\tilde{\mathbf{Z}}_t | \Omega_t^{\phi, q})^{\frac{1}{2}} \\ & \quad \text{Corr}(\phi(X_t), R | \Omega_t^{\phi, q}) \\ & \quad + \mathbb{E}_t[\phi(X) | \Omega_t^{\phi, q}] \mathbb{E}_t[R | \Omega_t^{\phi, q}]). \end{aligned} \quad (16)$$

Proof Appendix 4. ■

Here, $\text{Corr}(\phi(X_t), R | \Omega_t^{\phi, q})$ and $\mathbb{E}_t[R | \Omega_t^{\phi, q}]$ correspond to the correlation of $\phi(X_t)$ with stock returns and the average returns, respectively, within the q th quantile of $\phi(X_t)$. In consequence, maximizing $\text{Corr}(\phi(X_t), R_t)$ can be viewed as a multi-task learning and their coefficients determine the strength of different quantiles. Those coefficients are controllable through the marginal distribution of $\phi(X_t)$. That is, a larger variance of $\phi(X_t)$ within a given quantile tends to yield higher correlation with stock returns in that quantile.† Furthermore, it always holds that $\mathbb{E}_t[\phi(X) | \Omega_t^{\phi, q_2}] > \mathbb{E}_t[\phi(X) | \Omega_t^{\phi, q_1}]$ for $q_2 > q_1$, ensuring that (16) is rank-aware in the sense that learning tends to result in $\mathbb{E}_t[R | \Omega_t^{\phi, q_2}] > \mathbb{E}_t[R | \Omega_t^{\phi, q_1}]$. However, we can further increase the *Difference Gap*, defined as $\mathbb{E}_t[\phi(X) | \Omega_t^{\phi, q_2}] - \mathbb{E}_t[\phi(X) | \Omega_t^{\phi, q_1}]$, to make the q_2 th quantile on average more profitable.

To illustrate, we consider the case where only the first two moments of $\phi(X_t)$ remain invariant under Z-score standardization, while higher moments are unaddressed. Figure 6(a) presents the marginal distributions of $\phi(X_t)$ based on the ResNet1D architecture (Xu *et al.* 2025). These distributions appear nearly Gaussian but exhibit left skewness, supporting our intuition that *learning tends to capture non-implementable alphas from bottom quantiles*, as they exhibit larger variance and a greater Difference Gap in the lower quantiles. Figure 6(c) further corroborates this observation, showing that the bottom two quantiles outperform the others in terms of correlation. Additionally, for architectures with fewer parameters (i.e. lower expressibility) such as LSTM, capturing outliers may serve as a shortcut for learning. In such cases, the model tends to overfit to extreme stock return values while

neglecting the majority of samples. Figure 7(c) illustrates that the model primarily focuses on the two tails of $\phi(X_t)$ during training, with poor performance across the remaining quantiles. Furthermore, figure 7(a) shows that outliers appear in the marginal distributions on certain trading days, making these distributions hardly invariant. Notably, such a shortcut generally does not persist to the OOS domain, resulting in poor OOS performance for all quantiles. Consequently, the marginal distributions $\mathbb{P}_t(\phi(x))$ for all t should be aligned with a pre-defined distribution that satisfies the following conditions: (1) The distribution is right-skewed to prevent learning from being biased toward bottom quantiles, and (2) The distribution does not contain outliers at either tail.

Next, we propose a robust method to achieve distributional alignment between $\mathbb{P}_t(\phi(x))$ and a distribution with the desired shape. Specifically, firstly, we align $\phi'(X_t)$ with $U \sim \text{Uniform}(-1, 1)$ to prevent outliers, which leverages the desirable tail properties of the uniform distribution. We then derive ϕ using the transformation $\psi \circ \phi'$, followed by Z-score standardization to ensure a right-skewed distribution. Here, ψ is the smooth leaky ReLU function, defined as $\psi(u) := u\sigma(u) + 0.1u(1 - \sigma(u))$, where $\sigma(u) := \frac{1}{1+e^{-u}}$ is the sigmoid function. To accomplish this alignment, we minimize the Maximum Mean Discrepancy (MMD) (Tzeng *et al.* 2014) as follows:

$$\begin{aligned} \min_{\phi} \mathcal{L}_{\text{mmd}}^e(\phi) &:= \mathbb{E}_{\mathbb{P}_t \sim \hat{\mathbb{P}}_t} [\mathbb{E}_{X, X' \sim \mathbb{P}_t} [\kappa(\phi'(X), \phi'(X'))]] \\ & \quad + \mathbb{E}_{U, U' \sim \text{Uniform}(0,1)} [\kappa(U, U')] \\ & \quad - 2\mathbb{E}_{\substack{X \sim \mathbb{P}_t, \\ U \sim \text{Uniform}(0,1)}} [\kappa(\phi'(X), U)], \\ & \quad \forall e \in \{\text{hist}, \text{trg}\}, \end{aligned} \quad (17)$$

where $\kappa(\cdot, \cdot)$ denotes a customized kernel function. We adopt the Gaussian kernel $\kappa(x, y) := e^{-\frac{\|x-y\|^2}{0.2}}$, as it facilitates aligning all moments between the two distributions. Figures 6(b) and 7(b) demonstrate that our method yields $\mathbb{P}_t(\phi(x))$ with the desired shape. Furthermore, figure 6(d) shows that correlations in the top quantiles are more emphasized, while figure 7(d) confirms that the shortcut of learning from extremities is effectively mitigated.

3.4. Implementation of domain discovery

Differentiable Optimal Transport Plan. To solve (15) for α , we use the gradient descent method by treating the Wasserstein term $\mathcal{W}^2(\hat{\mathbb{P}}_{\text{src}}^{\alpha}(\mathbf{v}_t^{\phi}), \hat{\mathbb{P}}_{\text{trg}}(\mathbf{v}_t^{\phi}))$ as a neural network layer (figure 8), allowing gradients to propagate through. To achieve this, we adopt the DeepEMD framework (Zhang *et al.* 2022), which builds on the OptNet framework (Amos and Kolter 2017) and formulates linear or quadratic programming computations as differentiable optimization problems within neural networks. Since computing the Wasserstein distance for empirical distributions involves solving a linear program, DeepEMD utilizes the Implicit Function Theorem and Karush-Kuhn-Tucker (KKT) conditions to facilitate gradient backpropagation through the optimal transport solution.

Firstly, let $\mathbf{C} := (C_{ij})_{T+1 \leq i \leq T+T', 1 \leq j \leq T} \in \mathbb{R}^{|\mathcal{T}_{\text{hist}}| \times |\mathcal{T}_{\text{trg}}|}$ denote the transport cost matrix, where $C_{ij} := \|\mathbf{v}_{t-i}^{\phi} - \mathbf{v}_{t-j}^{\phi}\|^2$. Let

† $\text{Var}(\tilde{\mathbf{Z}}_t | \Omega_t^{\phi, q}) \rightarrow \text{Var}(\phi(X_t) | \Omega_t^{\phi, q})$ as $\phi(X_t)$ discovers $\tilde{\mathbf{Z}}$; we focus on $\tilde{\mathbf{F}} > 0$ as discussed in section 2.3.

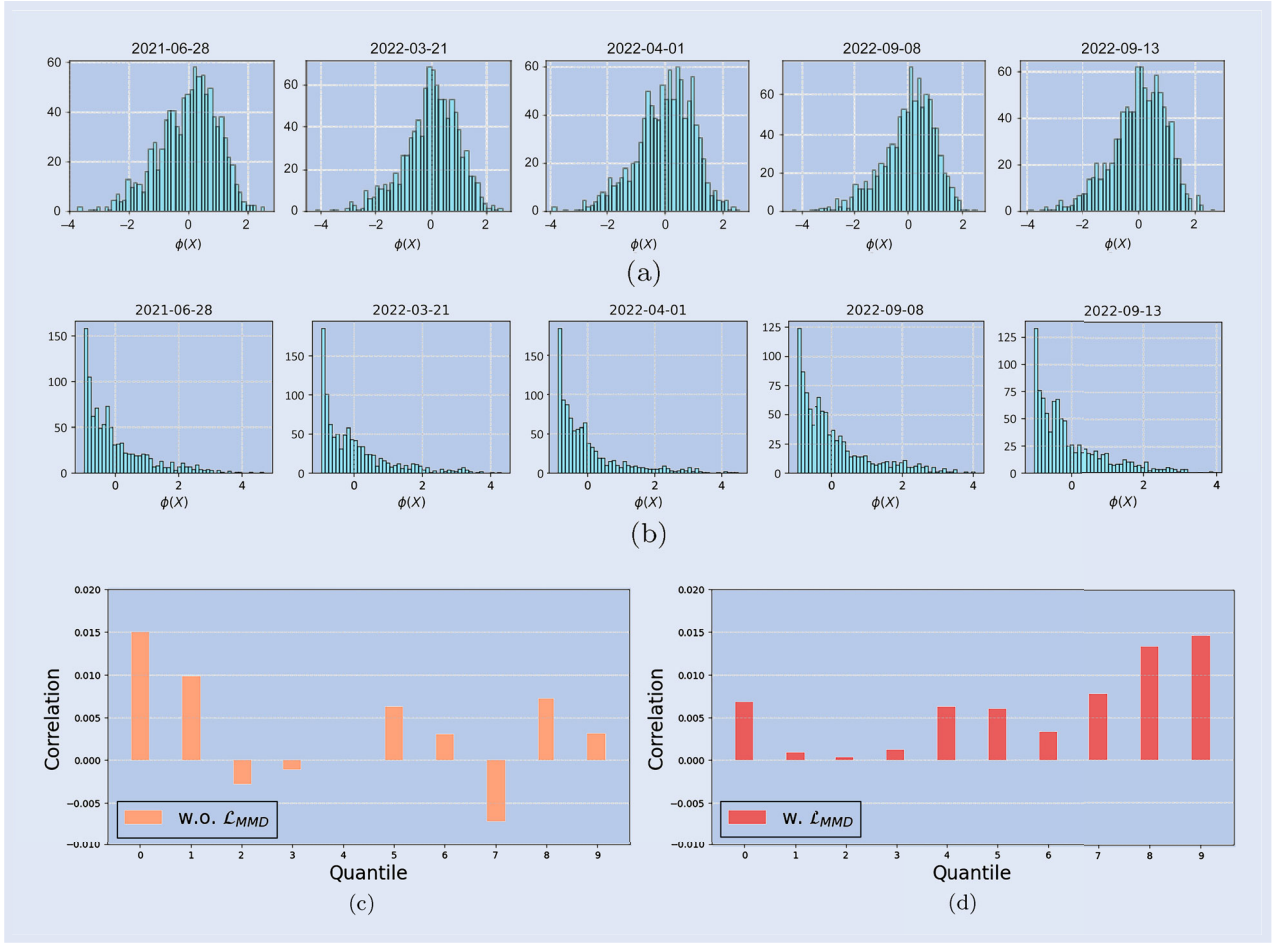


Figure 6. ϕ is parameterized by the ResNet1D architecture. (a) and (b) show in-sample histograms of $\phi(X_t)$ on five randomly selected trading days. (c) and (d) present estimates over the training horizon from December 22, 2020, to September 13, 2022, obtained by equally splitting $\phi(X_t)$ into ten quantiles on each trading day, computing correlations for each quantile, and averaging over the training horizon. (a) $\mathbb{P}_t(\phi(x))$; ResNet1D without $\mathcal{L}_{MMD}$. (b) $\mathbb{P}_t(\phi(x))$; ResNet1D with $\mathcal{L}_{MMD}$. (c) Quantile Correlations; ResNet1D without $\mathcal{L}_{MMD}$ and (d) Quantile Correlations; ResNet1D with $\mathcal{L}_{MMD}$.

$\pi := (\pi_{ij})_{T+1 \leq i \leq T+T', 1 \leq j \leq T} \in \mathbb{R}^{|\mathcal{T}_{hist}| \times |\mathcal{T}_{trg}|}$ be the transport plan. The optimal transport plan π^* is determined by solving the following linear program:

$$\min_{\pi \geq 0} \sum_{j=1}^T \sum_{i=T+1}^{T+T'} C_{ij} \pi_{ij}$$

subject to :

$$\begin{aligned} \sum_{j=1}^T \pi_{ij} &= \frac{\alpha_{t-i}}{\sum_{t \in \mathcal{T}_{hist}} \alpha_t} \quad \text{for } i = T+1, \dots, T+T', \\ \sum_{i=T+1}^{T+T'} \pi_{ij} &= \frac{1}{|\mathcal{T}_{trg}|} \quad \text{for } j = 1, \dots, T. \end{aligned} \quad (18)$$

Next, we define the Lagrangian function L for the optimization problem in equation (18). Let the dual variables $\lambda = (\lambda_i)_{T+1 \leq i \leq T+T'}$ and $\mu = (\mu_j)_{1 \leq j \leq T}$ correspond to the row and column marginal constraints, respectively, and let $\gamma = (\gamma_{ij})_{T+1 \leq i \leq T+T', 1 \leq j \leq T}$ enforce the non-negativity constraint $\pi_{ij} \geq 0$. The Lagrangian is expressed as:

$$L(\alpha, \pi, \lambda, \mu, \gamma)$$

$$\begin{aligned} &= \sum_{j=1}^T \sum_{i=T+1}^{T+T'} C_{ij} \pi_{ij} + \sum_{i=T+1}^{T+T'} \lambda_i \left(\frac{\alpha_{t-i}}{\sum_{t' \in \mathcal{T}_{hist}} \alpha_{t'}} - \sum_{j=1}^T \pi_{ij} \right) \\ &+ \sum_{j=1}^T \mu_j \left(\frac{1}{|\mathcal{T}_{trg}|} - \sum_{i=T+1}^{T+T'} \pi_{ij} \right) - \sum_{j=1}^T \sum_{i=T+1}^{T+T'} \gamma_{ij} \pi_{ij}. \end{aligned} \quad (19)$$

The KKT conditions are expressed in matrix form as:

$$f(\alpha, \pi, \lambda, \mu, \gamma) := \begin{bmatrix} \nabla_{\pi} L(\alpha, \pi, \lambda, \mu, \gamma) \\ \pi \mathbf{1} - \frac{\alpha}{\sum_{t' \in \mathcal{T}_{hist}} \alpha_{t'}} \\ \pi^{\top} \mathbf{1} - \frac{\mathbf{1}}{|\mathcal{T}_{trg}|} \\ \pi \odot \gamma \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix}, \quad (20)$$

where $\odot$ denotes the Hadamard product. For any α , let $(\pi^*, \lambda^*, \mu^*, \gamma^*)$ denote the corresponding zeros of (20). The *Implicit Function Theorem* (Theorems 9.1 and 9.2 in Munkres 2018) guarantees the existence of a neighborhood around α , denoted by B , and a differentiable function $\pi^* : B \rightarrow \mathbb{R}^{|\mathcal{T}_{hist}| \times |\mathcal{T}_{trg}|}$ such that $\forall \alpha \in B$, $f(\alpha, \pi^*(\alpha), \lambda^*, \mu^*, \gamma^*) = 0$, and the derivative of $\pi^*(\alpha)$

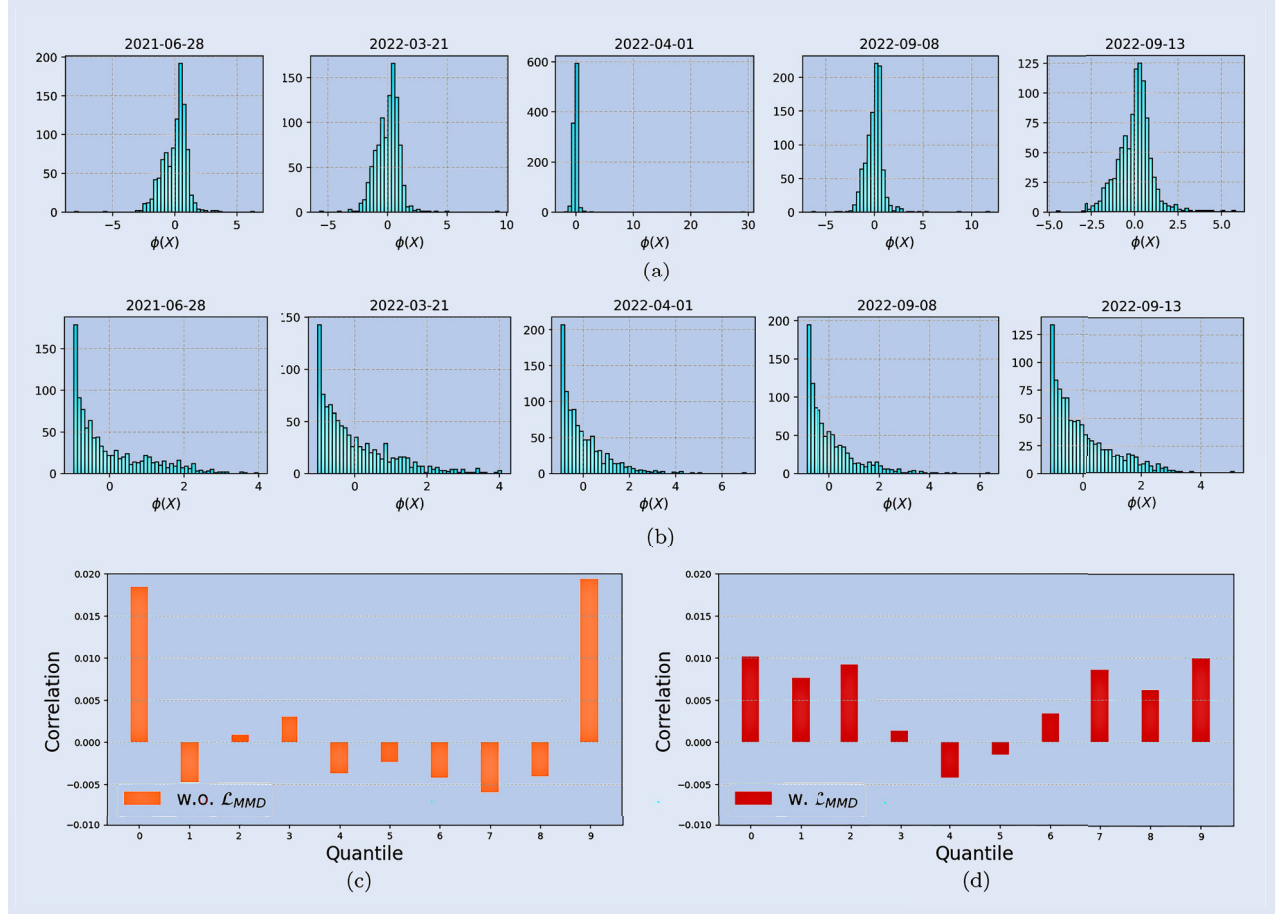


Figure 7. ϕ is parameterized by the architecture, LSTM. The figures are generated in the same way as in figure 6 on the training horizon between May 25, 2021 and February 15, 2023. (a) $\mathbb{P}_t(\phi(x))$; LSTM without $\mathcal{L}_{MMD}$. (b) $\mathbb{P}_t(\phi(x))$; LSTM with $\mathcal{L}_{MMD}$. (c) Quantile Correlations; LSTM without $\mathcal{L}_{MMD}$ and (d) Quantile Correlations; LSTM with $\mathcal{L}_{MMD}$.

has the explicit form:

$$\begin{aligned} D_\alpha \pi^*(\alpha) \\ = -D_\pi f(\alpha, \pi^*, \lambda^*, \mu^*, \gamma^*)^{-1} D_\alpha f(\alpha, \pi^*, \lambda^*, \mu^*, \gamma^*). \end{aligned} \quad (21)$$

As in figure 8, the Wasserstein distance $\mathcal{W}^2(\hat{\mathbb{P}}_{src}^\alpha(v_t^\phi), \hat{\mathbb{P}}_{trg}^\alpha(v_t^\phi))$ can be computed as: $\sqrt{\sum_{j=1}^T \sum_{i=T+1}^{T+T'} C_{ij} \pi_{ij}^*(\alpha)}$. The feedforward computation of $\pi^*(\alpha)$ is obtained by solving (18) by Sinkhorn Algorithm (Sinkhorn 1967). The backpropagation gradients during the training are computed using (21). As a result, the optimal transport plan can be directly treated as a common neural network layer.

Learnable α . To implement Domain Discovery, two tasks remain. First, each entry of α is binary-valued, so it needs to be made learnable. Second, the constraint in (15), $\sum_{t \in \mathcal{T}_{hist}} \alpha_t = T_\alpha$, must be satisfied. To achieve this, we encode α using the Gumbel-Softmax trick with a straight-through estimator (Jang *et al.* 2016). For each element α_t of α , the Gumbel-Softmax sample is computed as follows. We first sample Gumbel noise $v_{t'} := -\log(-\log(u_{t'}))$, where $u_{t'} \sim \text{Uniform}(0, 1)$ for $t' \in \mathcal{T}_{hist}$. Let ℓ_t represent the logits associated with each element α_t , and $\ell := (\ell_{t-T-1} \dots \ell_{t-T-T'})^\top$.[†]

[†] $u_{t'}$ is re-sampled for each batch during training.

Then, the continuous relaxation of α_t is given by:

$$\tilde{\alpha}_t(\ell) = \frac{e^{(\ell_t + v_t)/\tau}}{\sum_{t' \in \mathcal{T}_{hist}} e^{(\ell_{t'} + v_{t'})/\tau}}, \quad (22)$$

where $\tau > 0$ is a temperature parameter controlling the smoothness of the approximation. Since we aim to select $T_\alpha \gg 1$ micro-domains rather than the single most salient one, we set $\tau = 100$ to ensure a smoother approximation. Next, let $\tilde{\alpha} := (\tilde{\alpha}_{t-T-1} \dots \tilde{\alpha}_{t-T-T'})^\top$. The constraint in (15) is satisfied by setting $\tilde{\alpha}[T_\alpha]$ as the T_α th largest entry of $\tilde{\alpha}$ and defining α_t as:

$$\alpha_t = \mathbb{1}_{\tilde{\alpha}_t \geq \tilde{\alpha}[T_\alpha]}, \quad (23)$$

for $t \in \mathcal{T}_{hist}$, where $\mathbb{1}$ denotes the indicator function. Note that while (22) is differentiable with respect to ℓ , (23) is not. To address this, we use a straight-through estimator to make gradients passable:

$$\hat{\alpha}_t(\ell) = [\alpha_t(\ell) - \tilde{\alpha}_t(\ell)]_{stop} + \tilde{\alpha}_t(\ell), \quad (24)$$

where $[\cdot]_{stop}$ denotes stopping gradients. The resulting $\hat{\alpha} = (\hat{\alpha}_{t-T-1} \dots \hat{\alpha}_{t-T-T'})^\top \in \{0, 1\}^{|\mathcal{T}_{hist}|}$ follows the Gumbel Top- T_α distribution, with each entry's probability given by $\frac{e^{\ell_t}}{\sum_{t' \in \mathcal{T}_{hist}} e^{\ell_{t'}}$. Consequently, during training, we encode α using $\hat{\alpha}(\ell)$, and

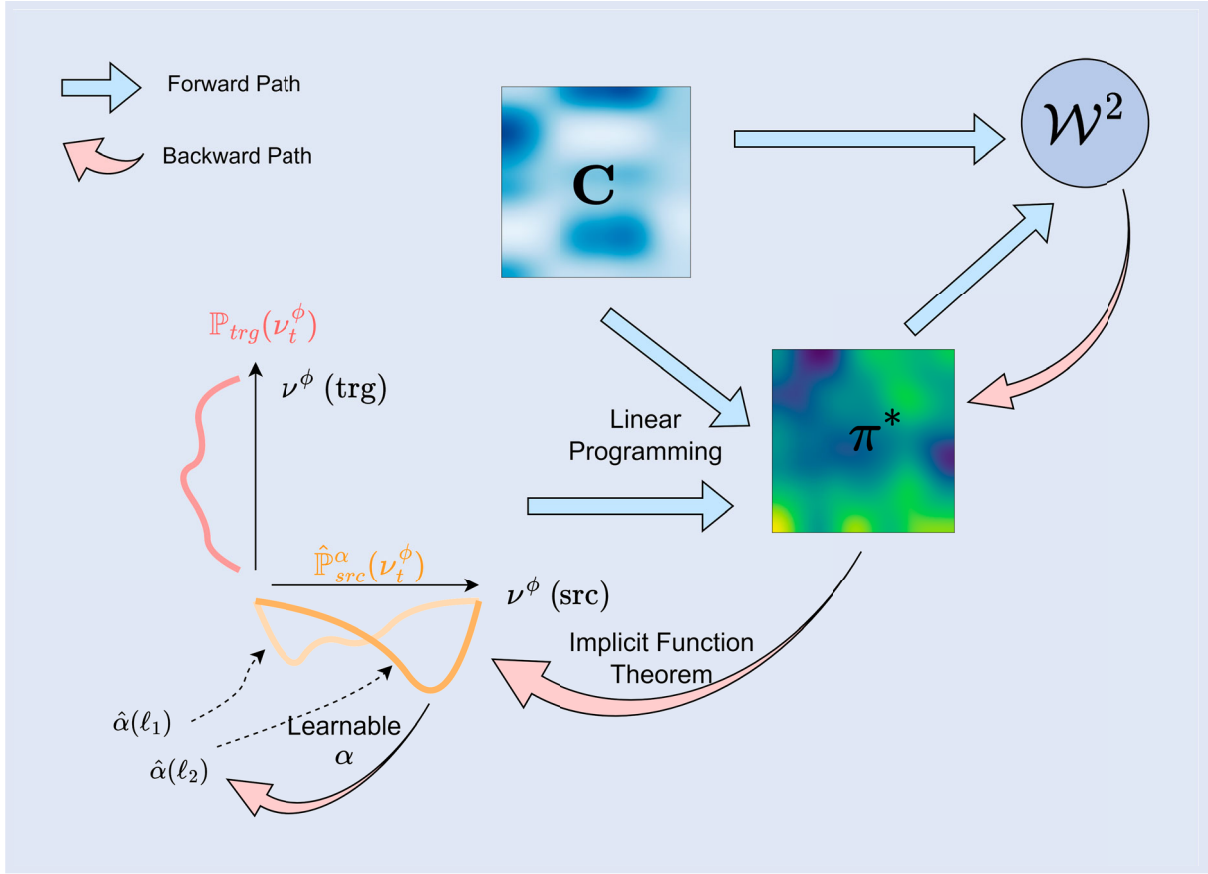


Figure 8. Overall architecture of the *Differentiable Optimal Transport Plan*. The blue arrows denote the forward computations, where the optimal transport plan is obtained by solving a linear programming problem defined by the cost matrix and boundary constraints. The red arrows indicate the backward propagation, where gradients flow through the transport plan with help of the Implicit Function Theorem and the learnable- α technique.

the learning objective corresponding to (15) is:

$$\begin{aligned} \min_{\ell \in \mathbb{R}^{|\mathcal{T}_{hist}|}} \mathcal{L}_{dd}(\ell; g_{trg}^*) \\ := 2\mathcal{W}^2(\hat{\mathbb{P}}_{src}^{\hat{\alpha}(\ell)}(\mathbf{v}_t^\phi), \hat{\mathbb{P}}_{trg}(\mathbf{v}_t^\phi)) \\ + \sqrt{\mathbb{E}_{\mathbf{v}_t^\phi \sim \hat{\mathbb{P}}_{src}^{\hat{\alpha}(\ell)}}[(\mathbf{v}_{trg}^* - \mathbf{v}_t^\phi)^2]} + \sqrt{1 - \mathbb{E}_{\mathbf{v}_t^\phi \sim \hat{\mathbb{P}}_{src}^{\hat{\alpha}(\ell)}}[(\mathbf{v}_t^\phi)^2]}. \end{aligned} \quad (25)$$

Here, $\mathcal{L}_{dd}(\ell; g_{trg}^*)$ explicitly specifies that ϕ is associated with g_{trg}^* . During inference, we compute $\hat{\alpha}(\ell)$ by setting $v_t = 0$ in (22) for all $t \in \mathcal{T}_{hist}$ to eliminate randomness.

3.5. Learning details

The training process consists of three stages. In the first stage, we train the prediction model on $\mathcal{T}_{trg}$ to estimate g_{trg}^* , which is utilized in the Domain Discovery Method (15). Additionally, during this stage, we ensure the prerequisites of (15) are satisfied, namely Z-score standardized $\phi(X)$ and invariant $\mathbb{P}_t(\phi(X)) \forall t \in \mathcal{T}_{hist} \cup \mathcal{T}_{trg}$. In the second stage, we employ the Domain Discovery Method to select relevant micro-domains from $\mathcal{T}_{hist}$. Finally, in the third stage, we continue training the prediction model using both the target domain data and the selected micro-domains

Causal Discovery Objectives. To reduce the Wasserstein term in either (10) or (11), the learning of g in the first

and third stages is guided by the objectives proposed in Xu et al. (2025), which implements the invariance-based Causal Discovery. Let $\mathcal{L}_{res}^e(\phi) := -\mathbb{E}_{\mathbf{v}_t^\phi \sim \hat{\mathbb{P}}_e}[(\mathbf{v}_t^\phi)^2]$, and let $\mathcal{L}_{inv}^e(\mathbf{v}, \phi) := \mathbb{E}_{\mathbf{v}_t^\phi \sim \hat{\mathbb{P}}_e}[(\mathbf{v}_t^\phi - \mathbf{v})^2]$. The overall Causal Discovery objective is then formulated as:

$$\begin{aligned} \min_{\phi, \mathbf{v}} \mathcal{L}_{cd}^e(\mathbf{v}, \phi) &:= \mathcal{L}_{res}^e(\phi) + \beta_1 \mathcal{L}_{inv}^e(\mathbf{v}, \phi) + \beta_2 \mathcal{L}_{mmd}^e(\phi), \\ \text{subject to } \mathbf{v}^2 &= -\mathcal{L}_{res}^e(\phi), \end{aligned} \quad (26)$$

where $\mathcal{L}_{mmd}^e(\phi)$ is defined in (17) and $\beta_1, \beta_2 > 0$ are hyper-parameters. We solve (26) using gradient descent on mini-batches. Each mini-batch, denoted by $\mathcal{T}_b$, contains fixed number of micro-domains and fixed number of stocks from each micro-domain. To handle the constraint in (26), the parameter $\mathbf{v}$ is updated iteratively. For the j th iteration, $\mathbf{v}$ is updated as:

$$\mathbf{v}_{j+1} = a\mathbf{v}_j + (1-a)\hat{\mathbf{v}}_b, \quad (27)$$

where $a > 0$ controls the momentum, and $\hat{\mathbf{v}}_b$ is computed as the square root of the batch-wise estimation of $-\mathcal{L}_{res}^e(\phi)$ (i.e. $\hat{\mathbf{v}}_b = \sqrt{\sum_{t \in \mathcal{T}_b} (\mathbf{v}_t^\phi)^2}$).

Stage 1. To satisfy the prerequisite of Domain Discovery Method, we define $\hat{\mathbb{P}}_{hist}$ such that $\mathbb{P}_t \sim \hat{\mathbb{P}}_{hist}$ for $t \in \mathcal{T}_{hist}$ are sampled uniformly. The learning objective for this stage is

formulated as:

$$\begin{aligned} \min_{\phi, \mathbf{v}} \mathcal{L}_{cd}^{trg}(\mathbf{v}, \phi) + \beta_2 \mathcal{L}_{mmd}^{hist}(\phi), \\ \text{subject to } \mathbf{v}^2 = -\mathcal{L}_{res}^{trg}(\phi). \end{aligned} \quad (28)$$

We solve (28) to obtain ϕ and $\mathbf{v}$, along with the target-only prediction model defined as $\hat{g}_{trg}(x) := \phi(x)\mathbf{v}$.

Stage 2. To reduce the estimation variance of $\hat{g}_{trg}$, we repeat the first stage M times and apply the Domain Discovery Method to the resulting collection of target-only models, denoted by $\{\hat{g}_{trg}^m\}_{m=1}^M$. The corresponding learning objective, based on (25) that ensures learnable α with respect to the parameters $\ell \in \mathbb{R}^{|\mathcal{T}_{hist}|}$, is formulated as:

$$\min_{\ell \in \mathbb{R}^{|\mathcal{T}_{hist}|}} \frac{1}{M} \sum_{m=1}^M \mathcal{L}_{dd}(\ell; \hat{g}_{trg}^m). \quad (29)$$

Using the resulting $\alpha(\ell)$, we then select the micro-domains, denoted by $\mathcal{T}_{src} := \{t \in \mathcal{T}_{hist} \mid \alpha_t(\ell) = 1\}$.

Stage 3. We continue the training on $\mathcal{T}_{src} \cup \mathcal{T}_{trg}$ from the models in $\{\hat{g}_{trg}^m\}_{m=1}^M$, with the learning objective:

$$\begin{aligned} \min_{\phi, \mathbf{v}} \mathcal{L}_{cd}^{trg}(\mathbf{v}, \phi) + \mathcal{L}_{cd}^{src}(\mathbf{v}, \phi), \\ \text{subject to } \mathbf{v}^2 = -\frac{\mathcal{L}_{res}^{trg}(\phi) + \mathcal{L}_{res}^{src}(\phi)}{2}. \end{aligned} \quad (30)$$

4. Experiments

Data. Our analysis is based on the Chinese A-share market from January 4, 2010, to October 16, 2024, covering over 4,000 stocks. We exclude stocks under two conditions: (1) *No Liquidity*, where the transaction amount is zero; and (2) *One-Price Trading Limit*, where the open, high, low, and close prices are identical. The out-of-sample (OOS) horizon spans September 15, 2022, to October 14, 2024, including 500 trading days, while the remaining days form the in-sample set. We focus on predicting daily returns based on open prices, calculated as:

$$r_d^n := \frac{\text{open}_{d+2}^n - \text{open}_{d+1}^n}{\text{open}_{d+1}^n}, \quad (31)$$

where open_d^n is the open price of stock n on day d . The input for each stock is structured as a 8×196 tensor, consisting of 8-day OHLCV data and signals from the *Alpha191* dataset, provided by GUOTAI JUNAN SECURITIES.[†] Min-max scaling is applied to the input data. For hypergraph architectures, we incorporate industry classification and stock concept data as in Xu *et al.* (2021).[‡] Each trading day's cross-sectional stock pool includes the top 1,000 largest companies

[†] OHLCV refers to Open, High, Low, Close, and Volume.

[‡] Data is sourced from Tushare: <https://tushare.pro>. Specifically, the stock concept data is retrieved via the API: *concept* and *concept_detail* in https://tushare.pro/document/1?doc_id=108; The industry data is retrieved via the API: *industry_classify* with level=L1 in https://tushare.pro/document/2?doc_id=181.

by market capitalization, ensuring no missing values in either the input data or returns.

Architectures. The following neural network architectures are considered:

- **LSTM** (Hochreiter and Schmidhuber 1997): Long Short-Term Memory (LSTM) networks are a type of recurrent neural network designed to effectively capture temporal dependencies in sequential data.
- **ResNet1D** (Xu *et al.* 2025): This refers to a Residual Network (He *et al.* 2016) with residual blocks based on 1-dimensional convolutional neural networks (LeCun *et al.* 1998). The convolutional blocks, commonly used for spatial data, are also effective for temporal data.
- **TFM** (Vaswani 2017): This refers to the Transformer encoder, a deep learning architecture that excels at modeling sequential data by leveraging self-attention mechanisms. It has demonstrated remarkable performance in large language models, revolutionizing the field of natural language processing.
- **HIST** (Xu *et al.* 2021): This architecture is specifically designed for stock trend forecasting by leveraging the pre-defined concepts, such as industries and main businesses, as well as mining hidden concepts using hypergraph techniques. The hypergraph layers extract cross-sectional stock information through stock-to-concept and concept-to-stock propagation.
- **CoarHGNN** (Yan *et al.* 2024): This architecture builds on the ideas of HIST but replaces the hypergraph layers with stacked hypergraph Laplacian layers (Bai *et al.* 2021). While HIST extracts cross-sectional stock information through neighboring nodes, CoarHGNN considers the entire graph information.
- **GAT** (Veličković *et al.* 2017): This refers to the Graph Attention Network, which extracts cross-sectional stock information using a stock-to-stock attention mechanism.

All graph-based architectures (i.e. HIST, CoarHGNN, and GAT) utilize LSTM as the feature encoder for individual stocks.

Portfolios, Settings and Evaluation Metrics. Portfolios are constructed to evaluate performance over the out-of-sample (OOS) period. Daily portfolio weights are determined by taking long positions in stocks based on $g(x)$. We consider two types of long positions:

- **full_port**: This portfolio invests in all stocks within the stock pool. The weight of each stock n is defined as $\frac{\text{rank}(g(x_d^n))}{\sum_{n=1}^N \text{rank}(g(x_d^n))}$, where $\text{rank}(g(x_d^n))$ denotes the descending rank of stock n 's model output on day d —that is, stocks with larger values of $g(x_d^n)$ are assigned higher ranks.
- **topP%_port**: This portfolio invests only in the top $P\%$ stocks within the stock pool. The weight of each stock n is defined as $\frac{\text{rank}_{P\%}(g(x_d^n))}{\sum_{n=1}^N \text{rank}_{P\%}(g(x_d^n))}$.

The term $\text{rank}_{P\%}(g(x_d^n))$ is defined as follows: Let τ_P denote the $(100-P)$ th percentile such that $\mathbb{P}_d(g(X) \geq \tau_P) = P$. Then, for stocks where $g(x_d^n) > \tau_P$, $\text{rank}_{P\%}(g(x_d^n))$ is the descending rank of $g(x_d^n)$ among the set $\{g(x_d^n) \mid g(x_d^n) > \tau_P\}$. For all other stocks (i.e. those not in the top $P\%$), $\text{rank}_{P\%}(g(x_d^n)) = 0$, meaning they are excluded from the portfolio.

To evaluate the *excess returns* attributable to our models— independent of general market movements—we hedge the long positions with a baseline portfolio. Specifically, we use a $1/N$ portfolio that equally weights all N stocks in the daily stock pool. This approach isolates the strategy’s alpha by neutralizing market-wide effects. These hedged portfolios are referred to as *full_excess_port* and *topP%_excess_port*, respectively.

In addition, we adopt the ranking-aware MSE loss (Sawhney et al. 2021) as the baseline learning objective function:

$$\begin{aligned} \min_g \mathcal{L}_{\text{base}}(g) := & \frac{1}{T} \sum_{d=1}^T \left(\sum_{n_2=1}^N \frac{1}{N} \sum_{n=1}^N \|g(x_{nd}) - r_{nd}\|^2 \right. \\ & + \beta_{\text{base}} \frac{1}{N^2} \sum_{n_1=1}^N \sum_{n_2=1}^N \max(0, -(g(x_{n_1d}) \\ & \left. - g(x_{n_2d}))(r_{n_1d} - r_{n_2d})) \right), \end{aligned} \quad (32)$$

which combines MSE with a regularization term that penalizes mis-ranked stocks. The regularization strength is controlled by β_{base} . For updating the prediction models, we consider the following schemes:

- *LongHOR*: This scheme follows the standard approach commonly used in the literature. Models are trained once using the closest 2050 trading days to the OOS horizon. The first 2000 days are used as the training set, and the subsequent 50 days serve as the validation set. The learning objective $\mathcal{L}_{\text{base}}$ is employed for model training.
- *Roll400(Base)*: Models are updated using baseline learning settings every 100 trading days, resulting in 5 updates during the OOS horizon. Training occurs on a rolling 450-day window, where the first 400 days form the training set and the remaining 50 days constitute the validation set. The learning objective $\mathcal{L}_{\text{base}}$ is applied.
- *RollD(CD)*: In this scheme, Causal Discovery is applied as defined in (26). The update frequency and train-validation split follow the same setup as Roll400(Base), but the length of the training horizon is varied, with $D \in \{100, 400, 700\}$.
- *DomDis*: This scheme incorporates our Domain Discovery Method. Similarly, the update frequency is every 100 trading days, with 5 updates during the OOS horizon. However, the in-sample domain for each update spans the most recent 2550 trading days. Within this period, the first 2400 days constitute the source domain, the next 100 days form the

target domain, and the remaining 50 days are used as the validation set.

For algorithms involving the Causal Discovery objective (26), each micro-domain $(t_{-i-1}, t_{-i}]$ is constructed from four consecutive trading days, and we denote $\mathcal{T}_{\text{tr}}$ and $\mathcal{T}_{\text{val}}$ as the sets of micro-domains for the training and validation sets, respectively. The models are optimized using the Adam optimizer (Kingma and Ba 2014), with each batch consisting of four micro-domains and 500 samples from each. The parameter $\mathbf{v}$ in (26) is initialized to 0.1. Training is terminated early if the average causal coefficients divided by their standard deviation on the validation set stagnate over ten successive epochs, except during **Stage 1** of DomDis, where training continues until convergence on the training set. For the Domain Discovery Method, we set $T_\alpha = 150$ (i.e. 600 trading days are selected from the entire in-sample horizon), and (29) is optimized using the Adam optimizer with a learning rate of 0.001 over 2000 iterations. We set $M = 4$, resulting in four final models after **Stage 3**. The learning process for Roll400(Base) and LongHOR uses the same batch size as other schemes, with each batch including all 1000 stocks from two random trading days. Early stopping is applied based on the value of $\mathcal{L}_{\text{base}}(g)$ on the validation set, and $\beta_{\text{base}}=0.2$ is used in (32). Additional hyperparameters for different architectures and schemes are detailed in Appendix 6.

Main Comparison. We train four models for each algorithm and report the averages and standard deviations for portfolio *full_excess_port* and *top10%_excess_port* in tables 2 and 3, respectively. Regarding *top10%_excess_port*, DomDis consistently outperforms its counterpart schemes in terms of average metric values for all architectures, except for the average SR of CoarHGNN, which nonetheless exhibits a significantly lower standard deviation compared to other schemes. Overall, DomDis achieves the lowest standard deviations across experiments, indicating reduced model variance when applying the Domain Discovery Method. Additionally, results from *full_excess_port* in table 3 confirm that the quantile-specific method does not compromise performance on other quantiles. DomDis continues to outperform other schemes, with exceptions for the average SR of LSTM and ResNet1D. Similarly, Roll400(CD), which also incorporates the quantile-specific method, generally performs better than schemes without such specialization. LongHOR and Roll400(Base) are commonly used in the literature. LongHOR assumes market stationarity and uses a very long training horizon to reduce noise during model learning, while Roll400(Base) balances nonstationarity and noise. For ResNet1D and TFM, it is critical to account for market nonstationarity, as their LongHOR configurations yield negative SR and CAGR. In contrast, GAT’s LongHOR outperforms its Roll400(Base) counterpart, reflecting the sensitivity of GAT’s self-attention mechanism to noise along the stock axis. For other architectures, Roll400(Base) generally enhances SR compared to LongHOR for *full_excess_port*. However, the SRs of both LongHOR and Roll400(Base) rarely exceed one. Furthermore, Roll400(CD) and DomDis, which employ the quantile-specific method, show significant improvements in CAGR from *full_excess_port* to *top10%_excess_port*, whereas no notable improvement is observed for LongHOR and Roll400(Base), implying that

Table 2. *top10%_excess_port*: Metrics are reported as the averages over 4 runs, with standard deviations shown in parentheses.

	SR				CAGR			
	LongHOR	Roll400 (Base)	Roll400 (CD)	DomDis	LongHOR	Roll400 (Base)	Roll400 (CD)	DomDis
LSTM	1.128 (0.293)	0.615 (0.508)	1.257 (0.199)	1.819 (0.140)	9.075% (2.538%)	4.401% (3.910%)	8.531% (1.637%)	18.594% (1.756%)
ResNet1D	− 0.252 (0.190)	0.238 (0.470)	1.426 (0.528)	1.600 (0.232)	− 2.602% (1.446%)	1.304% (3.246%)	8.661% (3.453%)	10.439% (1.756%)
TFM	0.016 (0.280)	0.418 (0.511)	1.475 (0.359)	1.727 (0.280)	0.593% (0.981%)	3.274% (4.030%)	12.768% (2.549%)	15.728% (2.336%)
HIST	0.601 (0.975)	0.549 (0.393)	1.163 (0.286)	1.610 (0.194)	4.743% (7.292%)	5.018% (3.880%)	9.855% (1.967%)	15.888% (2.114%)
CoarHGNN	0.177 (0.883)	0.240 (0.670)	1.733 (0.337)	1.574 (0.045)	3.982% (6.462%)	0.773% (3.522%)	14.855% (3.001%)	15.303% (0.366%)
GAT	0.549 (0.365)	0.213 (0.646)	1.371 (0.287)	1.666 (0.172)	4.744% (4.444%)	0.271% (4.122%)	10.616% (2.159%)	16.058% (1.755%)

Table 3. *full_excess_port*: Metrics are reported as the averages over 4 runs, with standard deviations shown in parentheses.

	SR				CAGR			
	LongHOR	Roll400 (Base)	Roll400 (CD)	DomDis	LongHOR	Roll400 (Base)	Roll400 (CD)	DomDis
LSTM	1.027 (0.252)	1.254 (0.576)	1.792 (0.087)	1.692 (0.205)	2.225% (0.445%)	2.171% (0.879%)	3.680% (0.128%)	5.901% (0.745%)
ResNet1D	− 0.008 (0.387)	0.984 (0.371)	1.531 (0.541)	1.499 (0.017)	− 0.070% (0.535%)	1.103% (0.467%)	2.926% (1.130%)	3.421% (0.072%)
TFM	− 0.163 (0.537)	0.978 (0.749)	1.378 (0.256)	1.808 (0.191)	− 0.499% (1.750%)	3.011% (2.191%)	3.280% (0.423%)	5.095% (0.586%)
HIST	0.185 (0.890)	0.461 (0.297)	1.437 (0.341)	1.561 (0.138)	0.410% (2.003%)	1.163% (0.787%)	3.906% (0.759%)	5.364% (0.485%)
CoarHGNN	0.674 (0.362)	0.752 (0.153)	1.505 (0.270)	1.730 (0.128)	1.924% (1.278%)	1.925% (0.408%)	4.054% (0.554%)	5.920% (0.448%)
GAT	1.035 (0.183)	0.430 (0.454)	0.767 (0.068)	1.423 (0.120)	3.363% (0.516%)	1.348% (1.327%)	1.444% (0.178%)	4.409% (0.393%)

Table 4. One-tailed p -values for paired t -tests with alternative hypothesis: DomDis outperforms Roll400(CD).

Model	top10%_excess_port		full_excess_port	
	SR	CAGR	SR	CAGR
LSTM	0.002	< 0.001	0.94	0.004
ResNet1D	0.17	0.05	0.55	0.22
TFM	0.10	0.03	0.02	0.002
HIST	0.02	0.003	0.24	0.01
CoarHGNN	0.86	0.22	0.09	0.001
GAT	0.04	0.001	< 0.001	< 0.001

the ranking-aware objective alone is insufficient to ensure superior top-quantile performance.

Table 4 reports the paired t -test results comparing DomDis against Roll400(CD) on both the *top10%_excess_port* and *full_excess_port* experiments. Each cell lists the t -statistic and the corresponding one-tailed p -value under the alternative hypothesis that DomDis outperforms Roll400(CD). For the *CAGR* metric, most p -values are below 0.05, indicating that DomDis achieves statistically significant improvements across the majority of architectures. In terms of *SR*, most results exhibit either statistically significant (< 0.05) or nearly significant (< 0.2) p -values, reflecting consistent and directionally stable improvements under the Domain Discovery framework. The slightly higher p -values observed for a few

models may result from the limited number of random seeds (four runs) and the inherent volatility of Sharpe Ratio estimates. Taken together with the strongly significant *CAGR* outcomes, these results confirm that DomDis enhances model performance in a statistically robust manner.

More Insights on DomDis. We conduct an ablation study on the Domain Discovery Method, comparing it with three counterparts: (1) Roll700(CD), (2) Roll100(CD), and (3) DomDis(Q1). Roll700(CD) uses the same number of samples as DomDis but arranges them in consecutive micro-domains. Roll100(CD) comprises 100 consecutive trading days, equivalent to the training horizon in **Stage 1** of DomDis. DomDis(Q1) represents the case where the Domain Discovery Method is applied, but the micro-domains are selected based on the lowest T_α entries of $\tilde{\alpha}(\ell)$, with Q1 referring to the bottom quantile of the selection scores, meaning the chosen micro-domains are the least suitable.

We train four models for all schemes and present the box-plots of *top10%_excess_port*’s *SR* and *CAGR* in figure 9. The sample sizes of Roll700(CD), DomDis(Q1), and DomDis are identical, with their only distinction being their exposure to market nonstationarity. DomDis(Q1) produces significantly worse results compared to all other schemes, providing strong evidence of the efficacy of our proposed Domain Discovery Method. Specifically, our methodology effectively identifies well-suited micro-domains while isolating those that substantially harm OOS performance. Furthermore, none

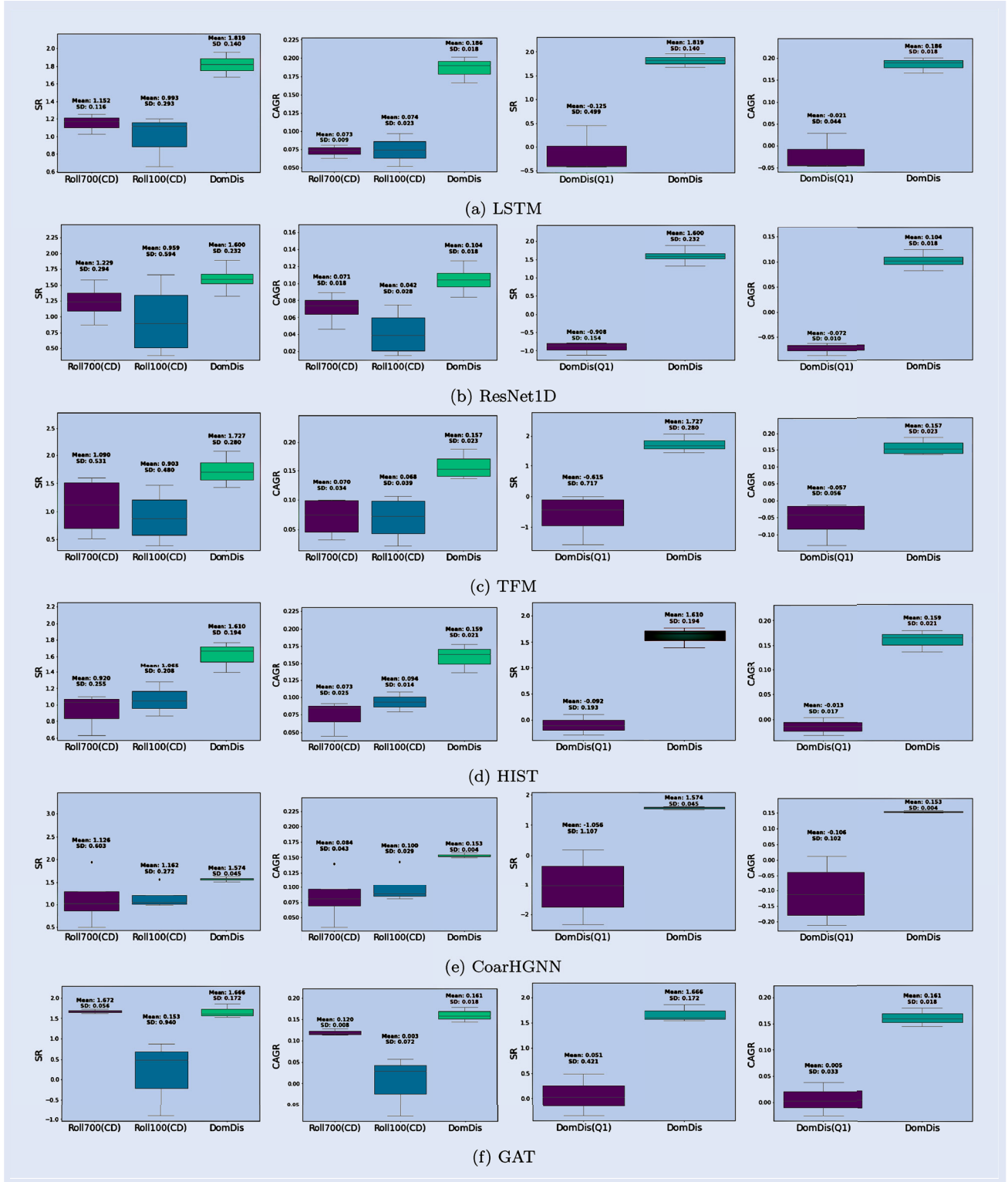


Figure 9. Boxplots of *top10%_excess_port*'s SR and CAGR for the schemes Roll700(CD), Roll100(CD), DomDis(Q1), and DomDis, with average and standard deviation values also provided. (a) ResNet1D. (b) TFM. (c) HIST. (d) CoarHGNN. (e) GAT and (f) LSTM.

of the Roll700(CD) results, except for GAT, are competitive with DomDis, highlighting that market nonstationarity negatively impacts the profitability of most architectures. As expected, Roll100(CD), which uses the smallest sample size, exhibits greater variance than both Roll700(CD) and DomDis. Notably, only HIST and CoarHGNN achieve higher average metric values with Roll100(CD) compared to Roll700(CD), despite Roll100(CD) being far less exposed to

market nonstationarity. This suggests that the sample size of Roll100(CD) is insufficient to achieve optimal performance.

For each architecture, we visualize the diversity of model outputs induced by all schemes. Specifically, we calculate the correlations among different model outputs for each trading day within the OOS horizon and compute their average across all OOS trading days. The distance matrix for the model outputs is then derived by subtracting these average

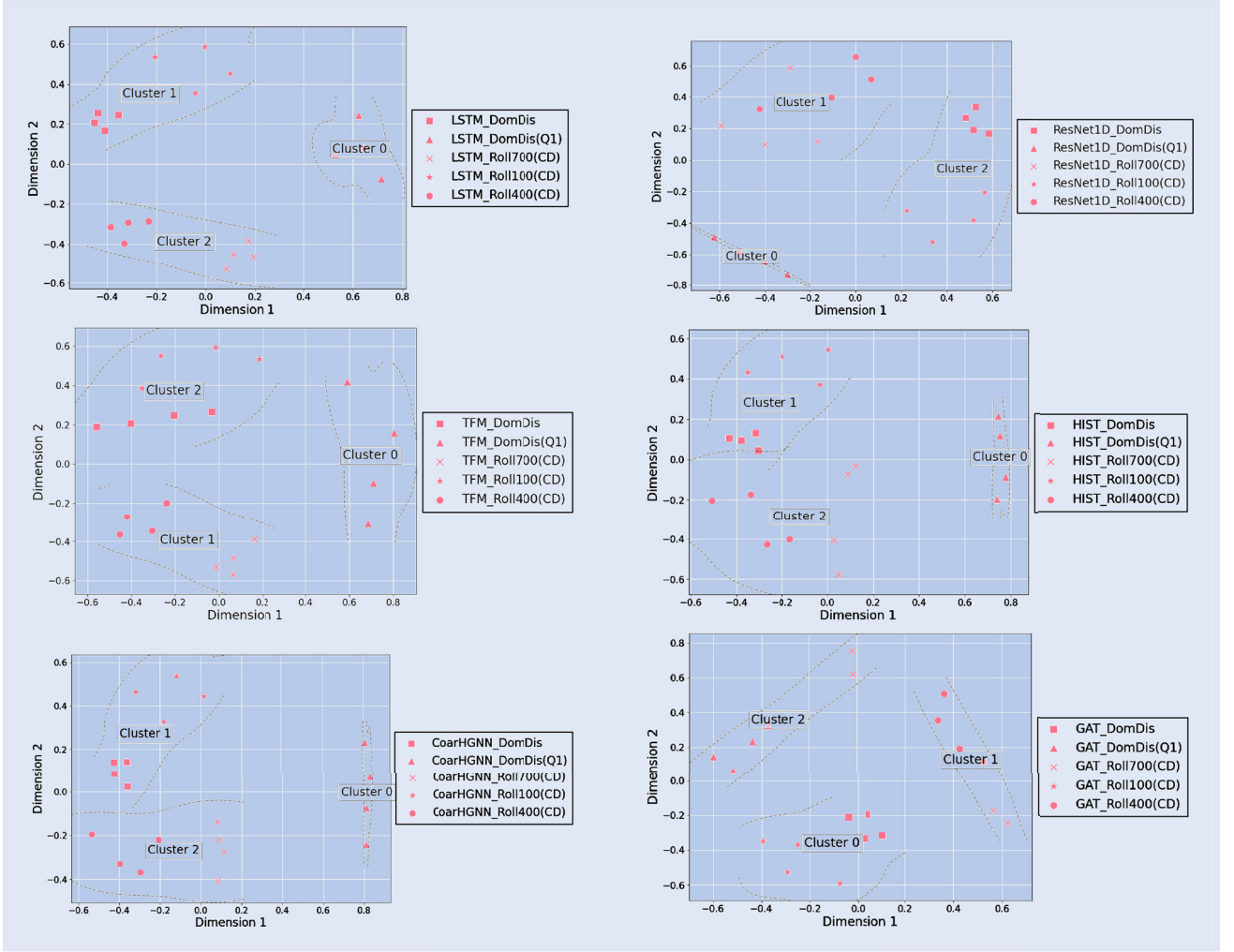


Figure 10. 2D visualization of the diversity of models across different schemes. Models for each architecture are clustered into three groups using K-Means.

correlations from one. In figure 10, the model outputs are embedded into a 2D space using Multidimensional Scaling (MDS) (Borg and Groenen 2007) applied to the distance matrix. Furthermore, we cluster the model outputs into three groups using K-Means (Hartigan and Wong 1979) for each architecture. The correlations of these clusters are shown in figure A1, derived by averaging model outputs within each cluster and computing the correlations. Interestingly, for all architectures, the model outputs of Roll100(CD) and DomDis belong to the same cluster, despite DomDis having a sample size seven times larger than Roll100(CD). This strongly supports the notion that our Domain Discovery Method surpasses the Noise-Nonstationarity Tradeoff. In addition, we observe that the models generated by Roll400, which are designed to balance noise and nonstationarity, closely resemble those produced by Roll700 while differing significantly from those by Roll100. This pattern is consistent across all neural network architectures. This finding suggests that the tradeoff in Roll400 fails to fully capture the alphas inherent to the most recent environment. As expected, DomDis(Q1) is distant from other schemes, further confirming its lack of effectiveness. Moreover, we observe that for the GAT architecture, Roll700(CD) and DomDis produce distinct model outputs, even though their average values of SR and $CAGR$ are similar,

as shown in figure 9. This highlights that Roll700(CD) cannot replace DomDis for the GAT architecture.

Top-Quantile Performance. We examine whether the quantile-specific method introduced in section 3.3 enhances top-quantile performance. In figure 11, we conduct an ablation study on Roll400(CD) by adding or removing the quantile-specific method and comparing the performance of portfolios $top50\%_port$, $top30\%_port$, $top20\%_port$, $top10\%_port$, $top5\%_port$, and $top2\%_port$. Our method consistently leads to increasing SR and $CAGR$ as investments are focused on higher percentiles of the stock bucket across all architectures. However, for the counterparts, this pattern is not guaranteed. Furthermore, the performance with our method significantly surpasses the counterparts with higher performance gap as higher percentiles are concentrated. To further validate the effectiveness of our approach, we plot the marginal distributions of $\phi(x)$ in figure 12 for the period between July 1, 2020 and July 9, 2024, showing that our method successfully achieves the desired right-skewed shape of the marginal distributions.

Hyperparameter's Sensitivity. We further conduct a sensitivity analysis on the optimization hyperparameters β_2 and the learning rate (lr). For each architecture, we run four independent trials by scaling the adopted hyperparameters of



Figure 11. OOS SR and CAGR for all architectures under the Roll400(CD) scheme, with and without the quantile-specific method. The portfolios *top50%_port*, *top30%_port*, *top20%_port*, *top10%_port*, *top5%_port*, and *top2%_port* are analyzed.

DomDis reported in Appendix 6 with a multiplicative factor $\rho \in \{0.2, 0.5, 0.8, 0.9, 1.0, 1.2, 2.0, 5.0\}$. The corresponding results are presented in figure 13. Overall, both β_2 and lr exhibit strong robustness when their values are perturbed within the range $\rho \in [0.5, 2]$, indicating that the proposed framework maintains stable performance without requiring precise hyperparameter tuning.

Trade Under Real Market Constraints. To emphasize the practical relevance of *Implementable Alpha*, we evaluate our strategy under real market constraints by explicitly accounting for both transaction costs (TC) and slippage. Specifically, we set $TC = 0.06\%$ for both the buy and sell sides. The evaluation focuses on the *top10%_excess_port*, where positions are updated using the Volume-Weighted Average Price (VWAP) between 9:30 am and 9:35 am to approximate realistic execution prices. For each architecture, we ensemble the model outputs of *DomDis* over four independent runs and apply a five-day rolling average to the ensembled alpha signals to obtain smoother predictions. These smoothed alphas help to regulate portfolio turnover and mitigate the impact of slippage. The performance results are reported in table 5, where both the SR and CAGR remain salient even after accounting for execution frictions, demonstrating that the alphas learned

Table 5. SR and CAGR of the smoothed ensembled signals from *DomDis*, evaluated after accounting for transaction costs (TC = 0.06% per side) and slippage.

	LSTM	ResNet1D	TFM	HIST	CoarHGNN	GAT
SR	1.21	1.13	1.26	1.48	1.37	1.47
CAGR	12.39%	3.34%	8.33%	9.39%	8.60%	9.26%

by our framework are not only statistically significant but also practically implementable in real trading environments.

5. Related topics

Machine-Learning Based Stock Prediction. Machine-learning techniques have increasingly been applied to stock market prediction, leveraging both time-series and cross-sectional data to identify patterns in stock returns. Deep learning architectures, such as Long Short-Term Memory (LSTM) networks (Hochreiter and Schmidhuber 1997) and Convolutional Neural Networks (CNNs) (He et al. 2016), have

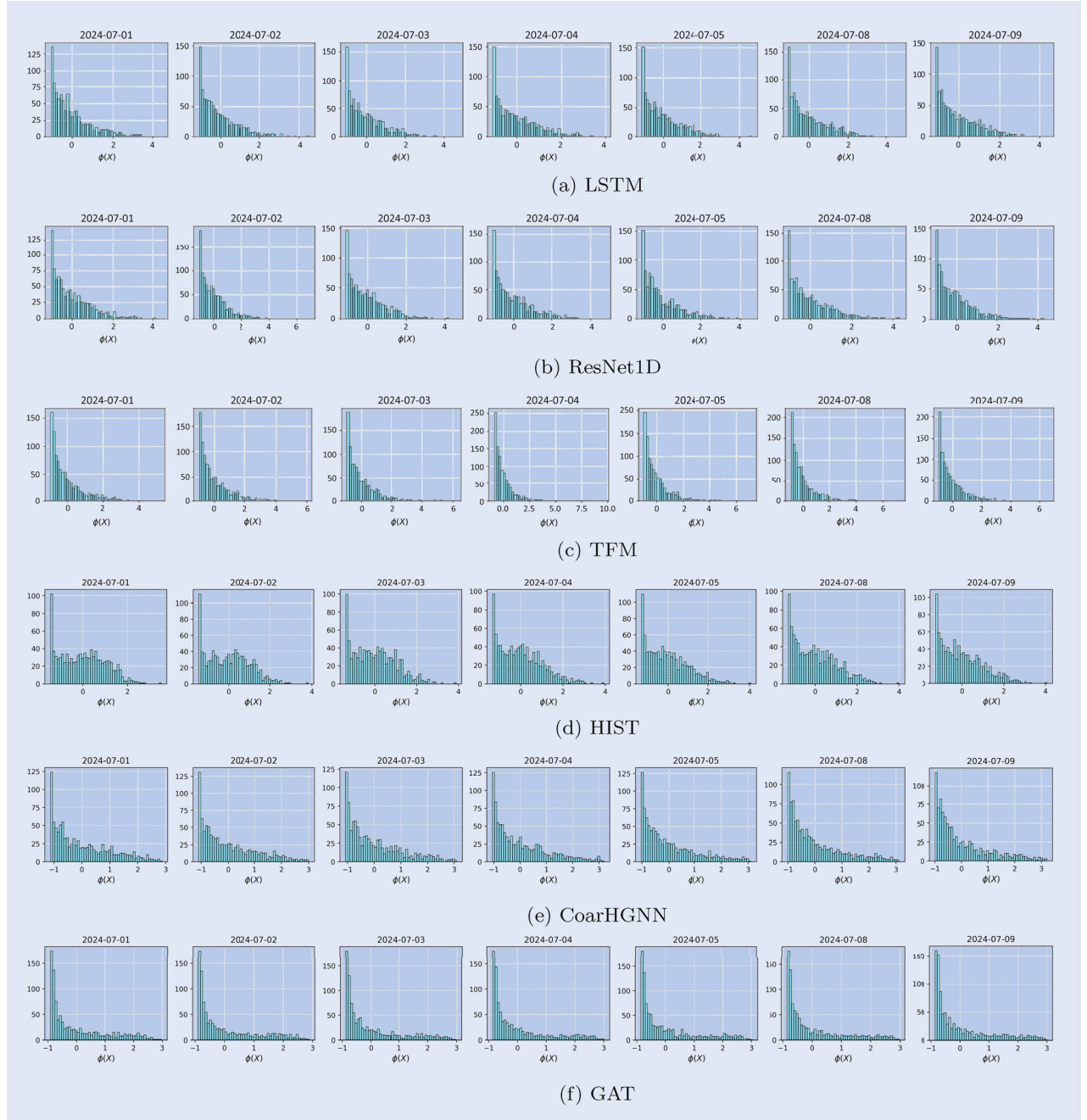


Figure 12. Marginal distributions of $g(x)$ for all architectures between 2024-07-01 and 2024-07-09 after applying the quantile-specific method. (a) LSTM. (b) ResNet1D. (c) TFM. (d) HIST. (e) CoarHGNN and (f) GAT.

been particularly popular for capturing temporal dependencies and nonlinearities inherent in stock data. Hybrid models that combine attention mechanisms and graph-based frameworks have also gained traction, as they facilitate nuanced information extraction across multiple financial indicators (Feng *et al.* 2019, Wang *et al.* 2021). Graph neural networks, such as HIST (Xu *et al.* 2021) and CoarHGNN (Yan *et al.* 2024), demonstrate enhanced prediction by modeling relationships between stocks. However, traditional machine-learning models struggle to maintain prediction accuracy in dynamic environments, particularly when confronted with high market noise and shifting correlations. This instability often limits model generalizability across varying economic cycles

and regimes. Jiang (2021) reviews 124 papers on successful applications of machine learning for stock market forecasting, highlighting these traits of the market as a persistent challenge for stock prediction.

Transfer Learning. Transfer Learning has become a pivotal technique in addressing tasks where available samples in the target domain is scarce, leveraging knowledge from a related source domain to improve learning outcomes (Pan and Yang 2009, Weiss *et al.* 2016). A key challenge in transfer learning is the discrepancy between the source and target domains, which can significantly affect the success of the transfer (Yosinski *et al.* 2014). Domain adaptation, a subfield of transfer learning, explicitly tackles this domain shift by

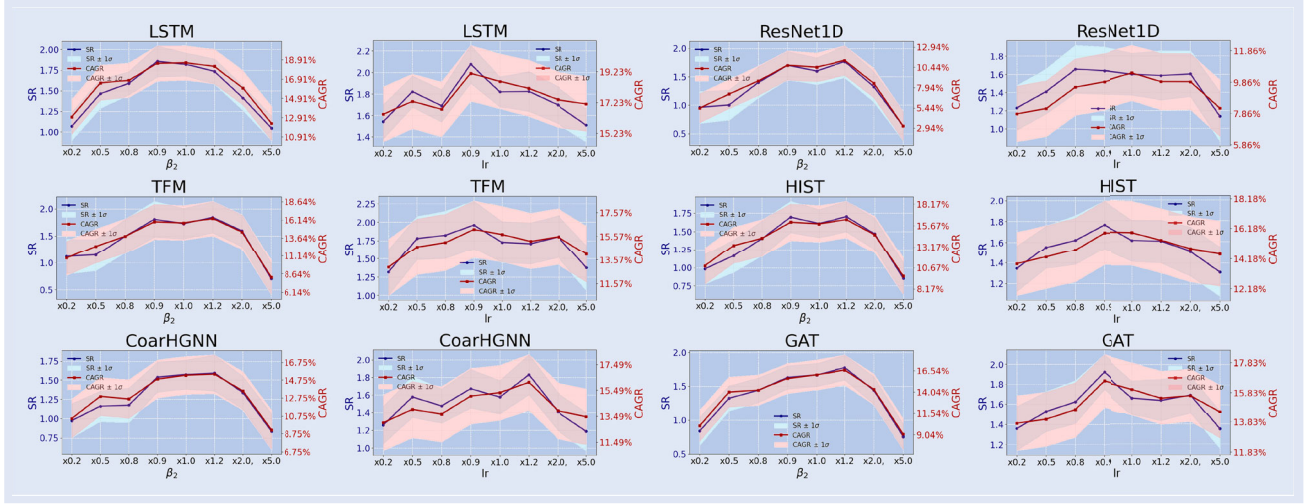


Figure 13. Tuning β_2 and the lr across architectures; Each subfigure shows the mean and $\pm 1\sigma$ range of SR and $CAGR$ for the *top10%_excess_port*.

learning models that generalize across domains (Ben-David *et al.* 2010). The theoretical foundation of domain adaptation is often anchored in the error bound of similar form of (3) that involves a term representing the ideal joint error (Mansour *et al.* 2009, Ben-David *et al.* 2010). When the ideal joint error is large, no hypothesis can simultaneously perform well on both domains, limiting the potential benefits of adaptation (Zhao *et al.* 2019, Sicilia *et al.* 2023). The ideal joint error reflects the feasibility of domain adaptation, while it is generally not manageable.

Market Nonstationarity. Nonstationarity presents a significant challenge to model robustness, as the performance of a model trained in one market condition may not transfer effectively to another. Baldovin *et al.* (2011) analyzes five years of daily high-frequency EUR/USD trading data and identifies the nonstationarity as the key statistical patterns, suggesting the standard models may fall short due to their reliance on stationary assumptions. Tu (2010) proposes a Bayesian framework that incorporate regime-switching dynamics when making portfolio choices. This approach is compared to traditional single-state models, highlighting how ignoring regime changes can lead to significant economic costs for investors. Even the Mean-Variance Optimization (Markowitz 1952), the Nobel-prize level work, struggles with nonstationary conditions, and Keller *et al.* (2015) mitigates the issue by re-optimizing the model using short lookback periods in a rolling manner, evidencing the existence of short-period momentum of market features. Indeed, the short-period momentum observed in stock markets is effectively leveraged in machine learning-based models in order to alleviate the influence of market nonstationarity. By regenerating models in a rolling manner, as demonstrated in (Bao *et al.* 2017, Zhou *et al.* 2018, Kim *et al.* 2019, Nguyen and Yoon 2019, Wang *et al.* 2021), the models dynamically incorporate the latest market data, enabling them to adapt to rapidly changing conditions. Xu *et al.* (2025) links the ideal joint error with the market nonstationarity, arguing that market's short-period momentum is likely to result in a small ideal joint error so that the discrepancy between the in-sample and OOS environments can be limited.

6. Discussion

This study presents a novel Domain Discovery framework designed to address the challenges of nonstationarity in stock market prediction, introducing a cutting-edge methodology grounded in advancements from the machine learning community. By leveraging causal relationships, our approach effectively expands the pool of labeled data while maintaining robustness to dynamic market conditions, offering a tailored solution to stock prediction challenges. The proposed framework places an innovative emphasis on top-quantile performance, aligning model priorities with practical investment objectives to optimize returns from the most profitable stocks. Extensive experiments validate the efficacy and versatility of our method in navigating complex market dynamics, highlighting the transformative potential of machine learning in financial modeling.

Since the primary contributions of this work stem from rigorous theoretical derivations, the proposed methodologies and their implementations are developed in a theoretically sound manner to ensure consistency with the underlying formulations. Nevertheless, exploring heuristic algorithms that simplify implementation or improve computational efficiency represents a promising future direction. For example, our Domain Discovery process can be viewed as selecting a representative subset of samples from the source domains that best match the target domain. This resemblance indicates that future research could take inspiration from heuristic subsampling approaches, such as those proposed in Zhang *et al.* (2023), to develop practical alternatives to the current *Differentiable Optimal Transport* component, offering a balance between theoretical rigor and computational complexity.

In addition, the Domain Discovery framework may face limitations in scenarios where the market environment undergoes rapid shifts such that the core assumption that *the most recent market environment is similar to the near future* is violated. A potential solution involves replacing the target domain with one that is better aligned to the forthcoming market environment. Future research will focus on parameterizing

and identifying new market environment and constructing target domains tailored to these regimes.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Appendix 1. Proof of Proposition 3.1

For $e \in \{src, trg\}$, let g_e^* be the minimizer with respect to the joint error $\min_{g \in \Omega^g} \mathbf{err}_e(g) + \mathbf{err}_{t_0}(g)$. Then,

$$\begin{aligned}
 \mathbf{err}_{t_0}(g) &= \int \ell(g(x_{t_0}), r_{t_0}) d\mathbb{P}_{t_0}(x_{t_0}, r_{t_0}) \\
 &\leq \int \ell(g(x_{t_0}), g_e^*(x_{t_0})) d\mathbb{P}_{t_0} + \int \ell(g_e^*(x_{t_0}), r_{t_0}) d\mathbb{P}_{t_0} \\
 &\leq \int \ell(g(x_{t_0}), g_e^*(x_{t_0})) d\mathbb{P}_{t_0} - \int \ell(g_e^*(x_e), r_e) d\hat{\mathbb{P}}_e \\
 &\quad + \int \ell(g_e^*(x_e), r_e) d\hat{\mathbb{P}}_e + \int \ell(g_e^*(x_{t_0}), r_{t_0}) d\mathbb{P}_{t_0} \\
 &\leq \left| \int \ell(g(x_{t_0}), g_e^*(x_{t_0})) d\mathbb{P}_{t_0} - \int \ell(g(x_e), g_e^*(x_e)) d\hat{\mathbb{P}}_e \right| \\
 &\quad + \left| \int \ell(g(x_e), g_e^*(x_e)) d\hat{\mathbb{P}}_e - \int \ell(g_e^*(x_e), r_e) d\hat{\mathbb{P}}_e \right| + \lambda_e^* \\
 &\leq \underbrace{\left| \int \ell(g(x_{t_0}), g_e^*(x_{t_0})) d\mathbb{P}_{t_0} - \int \ell(g(x_e), g_e^*(x_e)) d\hat{\mathbb{P}}_e \right|}_{(\star)} \\
 &\quad + \mathbf{err}_e(g) + \lambda_e^*. \tag{A1}
 \end{aligned}$$

Let $\pi \in \Pi$ where Π is the set of all couplings between $\hat{\mathbb{P}}_e(\cdot)$ and $\mathbb{P}_{t_0}(\cdot)$, and then Kantorovitch-Rubinstein theorem (Villani 2009) indicates that:

$$\begin{aligned}
 (\star) &\leq \int |\ell(g(x_{t_0}), g_e^*(x_{t_0})) - \ell(g(x_e), g_e^*(x_e))| d\pi \\
 &\leq \int |\ell(g(x_{t_0}), g_e^*(x_{t_0})) - \ell(g(x_e), g_e^*(x_{t_0}))| \\
 &\quad + |\ell(g(x_e), g_e^*(x_{t_0})) - \ell(g(x_e), g_e^*(x_e))| d\pi \\
 &\leq \int \ell(g(x_e), g(x_{t_0})) + \ell(r_e, r_{t_0}) d\pi + \lambda_e^*. \tag{A2}
 \end{aligned}$$

The above inequality is satisfied for all choices of $\pi \in \Pi$. Therefore,

$$\begin{aligned}
 (\star) &\leq \inf_{\pi \in \Pi} \int \ell(g(x_e), g(x_{t_0})) + \ell(r_e, r_{t_0}) d\pi + \lambda_e^* \\
 &= \mathcal{W}^1(\hat{\mathbb{P}}_e(g(x), r), \mathbb{P}_{t_0}(g(x), r)) + \lambda_e^*. \tag{A3}
 \end{aligned}$$

We complete the proof by combining (A1) and (A3).

Appendix 2. Proof of Proposition 3.2

Let $g_{src}^* \in \arg \min_{g \in \Omega^g} \mathbf{err}_{src}(g) + \mathbf{err}_{t_0}$. Then,

$$\begin{aligned}
 \lambda_{src}^* &= \int l(g_{src}^*(x), r) d\hat{\mathbb{P}}_{src}^\alpha + \int l(g_{src}^*(x), r) d\mathbb{P}_{t_0} \\
 &\leq \int l(g_{trg}^*(x), r) d\hat{\mathbb{P}}_{src}^\alpha + \int l(g_{trg}^*(x), r) d\mathbb{P}_{t_0} \\
 &= \int l(g_{trg}^*(x), r) d\hat{\mathbb{P}}_{src}^\alpha - \int l(g_{trg}^*(x), r) d\mathbb{P}_{trg} \\
 &\quad + \int l(g_{trg}^*(x), r) d\hat{\mathbb{P}}_{trg} + \int l(g_{trg}^*(x), r) d\mathbb{P}_{t_0} \\
 &= \int l(g_{trg}^*(x), r) (d\hat{\mathbb{P}}_{src}^\alpha - d\mathbb{P}_{trg}) \\
 &\quad + \lambda_{trg}^* \quad (\text{By Assumption 3.1}) \tag{A4}
 \end{aligned}$$

Let $\pi \in \Pi$ where Π is the set of all couplings between $\hat{\mathbb{P}}_{src}^\alpha$ and $\hat{\mathbb{P}}_{trg}$, and then Kantorovitch-Rubinstein theorem (Villani 2009) indicates that:

$$\begin{aligned}
 (\star) &\leq \int |l(g_{trg}^*(x_{src}), r_{src}) \\
 &\quad - l(g_{trg}^*(x_{trg}), r_{trg})| d\pi(x_{src}, x_{trg}, r_{src}, r_{trg}) \\
 &\leq \int |l(g_{trg}^*(x_{src}), r_{src}) - l(g_{trg}^*(x_{trg}), r_{src})| \\
 &\quad + |l(g_{trg}^*(x_{trg}), r_{src}) - l(g_{trg}^*(x_{trg}), r_{trg})| d\pi \\
 &\leq \int l(g_{trg}^*(x_{src}), g_{trg}^*(x_{trg})) + l(r_{src}, r_{trg}) d\pi \tag{A5}
 \end{aligned}$$

The above inequality is satisfied for all choices of $\pi \in \Pi$. Therefore,

$$\begin{aligned}
 \lambda_{src}^* &\leq \min_{\pi \in \Pi} \int l(g_{trg}^*(x_{src}), g_{trg}^*(x_{trg})) + l(r_{src}, r_{trg}) d\pi + \lambda_{trg}^* \\
 &= \mathcal{W}^1(\mathbb{P}_{src}(g_{trg}^*(x), r), \mathbb{P}_{trg}(g_{trg}^*(x), r)) + \lambda_{trg}^*. \tag{A6}
 \end{aligned}$$

Appendix 3. Proof of efficient alternative

By definition, $g_{trg}^*(x)$ can be written as $g_{trg}^*(x) = \phi(x)\mathbf{v}$. By Lemma A.1, we have $\mathbf{v} = \mathbb{E}_{\mathbf{v}_t^\phi \sim \hat{\mathbb{P}}_{trg}^\alpha} [\mathbf{v}_t^\phi]$. Also, suppose $\mathbb{E}_t[\phi(X)^2] = 1$ and $\mathbb{P}_t(\phi(x)) = \mathbb{P}_{t'}(\phi(x)) \forall \mathbb{P}_t, \mathbb{P}_{t'} \sim \hat{\mathbb{P}}_{src}^\alpha$ or $\hat{\mathbb{P}}_{trg}$. Let π^* be the optimal coupling with respect to $\mathcal{W}^1(\hat{\mathbb{P}}_{src}^\alpha(g_{trg}^*(x), r), \hat{\mathbb{P}}_{trg}(g_{trg}^*(x), r))$. Also, let π^{**} be the coupling such that $\pi^{**}(x_{src}, x_{trg} | \mathbb{P}_t, \mathbb{P}_{t'})$ is the optimal coupling with respect to $\mathcal{W}^1(\mathbb{P}_t(\phi(x)), \mathbb{P}_{t'}(\phi(x))) \forall \mathbb{P}_t, \mathbb{P}_{t'} \sim \hat{\mathbb{P}}_{src}^\alpha$ or $\hat{\mathbb{P}}_{trg}$, and $\pi^{**}(\mathbf{v}_{src}^\phi, \mathbf{v}_{trg}^\phi)$ is the optimal coupling with respect to $\mathcal{W}^2(\hat{\mathbb{P}}_{src}^\alpha(\mathbf{v}_t^\phi), \hat{\mathbb{P}}_{trg}(\mathbf{v}_t^\phi))$.

Then,

$$\begin{aligned}
 &\mathcal{W}^1(\hat{\mathbb{P}}_{src}^\alpha(g_{trg}^*(x), r), \hat{\mathbb{P}}_{trg}(g_{trg}^*(x), r)) \\
 &= \int |g_{trg}^*(x_{src}) - g_{trg}^*(x_{trg})| + |r_{src} - r_{trg}| d\pi^*
 \end{aligned}$$

$$\begin{aligned}
&= \int |\phi(x_{src})\mathbf{v} - \phi(x_{trg})\mathbf{v}| + |r_{src} - r_{trg}| d\pi^* \\
&\leq \int |\phi(x_{src})\mathbf{v} - \phi(x_{trg})\mathbf{v}| + |r_{src} - r_{trg}| d\pi^{**} \\
&\leq \int |\phi(x_{src})\mathbf{v} - \phi(x_{src})\mathbf{v}_{src}^\phi| + |\phi(x_{src})\mathbf{v}_{src}^\phi - \phi(x_{trg})\mathbf{v}_{trg}^\phi| \\
&\quad + |\phi(x_{trg})\mathbf{v}_{trg}^\phi - \phi(x_{trg})\mathbf{v}| \\
&\quad + |r_{src} - \phi(x_{src})\mathbf{v}_{src}^\phi| + |\phi(x_{src})\mathbf{v}_{src}^\phi - \phi(x_{trg})\mathbf{v}_{trg}^\phi| \\
&\quad + |\phi(x_{trg})\mathbf{v}_{trg}^\phi - r_{trg}| d\pi^{**} \\
&= \int |\phi(x)(\mathbf{v} - \mathbf{v}_t^\phi)| d(\hat{\mathbb{P}}_{src}^\alpha + \hat{\mathbb{P}}_{trg}) \\
&\quad + \int |r - \phi(x)\mathbf{v}_t^\phi| d(\hat{\mathbb{P}}_{src}^\alpha + \hat{\mathbb{P}}_{trg}) \\
&\quad + 2 \int |\phi(x_{src})\mathbf{v}_{src}^\phi - \phi(x_{trg})\mathbf{v}_{trg}^\phi| d\pi^{**} \\
&\leq \left(\int \phi(x)^2 d\hat{\mathbb{P}}_{src}^\alpha \right)^{\frac{1}{2}} \left(\int (\mathbf{v} - \mathbf{v}_t^\phi)^2 d\hat{\mathbb{P}}_{src}^\alpha \right)^{\frac{1}{2}} \\
&\quad + \left(\int \phi(x)^2 d\hat{\mathbb{P}}_{trg} \right)^{\frac{1}{2}} \left(\int (\mathbf{v} - \mathbf{v}_t^\phi)^2 d\hat{\mathbb{P}}_{trg} \right)^{\frac{1}{2}} \\
&\quad + \left(\int (r - \phi(x)\mathbf{v}_t^\phi)^2 d\hat{\mathbb{P}}_{src}^\alpha \right)^{\frac{1}{2}} + \left(\int (r - \phi(x)\mathbf{v}_t^\phi)^2 d\hat{\mathbb{P}}_{trg} \right)^{\frac{1}{2}} \\
&\quad + 2 \int |\phi(x_{src})\mathbf{v}_{src}^\phi - \phi(x_{trg})\mathbf{v}_{trg}^\phi| d\pi^{**} \\
&= \sqrt{\mathbb{E}_{src}[(\mathbf{v} - \mathbf{v}_t^\phi)^2]} + \sqrt{\mathbb{E}_{trg}[(\mathbf{v} - \mathbf{v}_t^\phi)^2]} \\
&\quad + \sqrt{1 - \mathbb{E}_{src}[(\mathbf{v}_t^\phi)^2]} + \sqrt{1 - \mathbb{E}_{trg}[(\mathbf{v}_t^\phi)^2]} \\
&\quad + 2 \underbrace{\int |\phi(x_{src})\mathbf{v}_{src}^\phi - \phi(x_{trg})\mathbf{v}_{trg}^\phi| d\pi^{**}}_{(*)}. \tag{A7}
\end{aligned}$$

After that,

$$\begin{aligned}
(*) &\leq \int |\phi(x_{src})\mathbf{v}_{src}^\phi - \phi(x_{trg})\mathbf{v}_{src}^\phi| \\
&\quad + |\phi(x_{trg})\mathbf{v}_{src}^\phi - \phi(x_{trg})\mathbf{v}_{trg}^\phi| d\pi^{**} \\
&\leq \int |\mathbf{v}_{src}^\phi| \left(\int |\phi(x_{src}) - \phi(x_{trg})| d\pi^{**}(x_{src}, x_{trg} | \mathbb{P}_t, \mathbb{P}_{t'}) \right) \\
&\quad d\pi^{**}(\mathbf{v}_{src}^\phi, \mathbb{P}_t, \mathbb{P}_{t'}) \\
&\quad + \left(\int \phi(x)^2 d\mathbb{P}_{trg} \right)^{\frac{1}{2}} \left(\int (\mathbf{v}_{src}^\phi - \mathbf{v}_{trg}^\phi)^2 d\pi^{**} \right)^{\frac{1}{2}} \\
&= \int |\mathbf{v}_{src}^\phi| \underbrace{\mathcal{W}^1(\mathbb{P}_t(\phi(x), \mathbb{P}_{t'}(\phi(x))))}_{=0} d\pi^{**}(\mathbf{v}_{src}^\phi, \mathbb{P}_t, \mathbb{P}_{t'}) \\
&\quad + \mathcal{W}^2(\hat{\mathbb{P}}_{src}^\alpha(\mathbf{v}^\phi), \hat{\mathbb{P}}_{trg}(\mathbf{v}^\phi)) \\
&= \mathcal{W}^2(\hat{\mathbb{P}}_{src}^\alpha(\mathbf{v}^\phi), \hat{\mathbb{P}}_{trg}(\mathbf{v}^\phi))
\end{aligned}$$

To combine the above, we have:

$$\begin{aligned}
&\mathcal{W}^1(\hat{\mathbb{P}}_{src}^\alpha(g_{trg}^*(x), r), \hat{\mathbb{P}}_{trg}(g_{trg}^*(x), r)) \\
&\leq 2\mathcal{W}^2(\hat{\mathbb{P}}_{src}^\alpha(\mathbf{v}_t^\phi), \hat{\mathbb{P}}_{trg}(\mathbf{v}_t^\phi)) + \sum_{e \in \{src, trg\}} \sqrt{\mathbb{E}_e[(\mathbf{v} - \mathbf{v}_t^\phi)^2]} \\
&\quad + \sum_{e \in \{src, trg\}} \sqrt{1 - \mathbb{E}_e[(\mathbf{v}_t^\phi)^2]}. \tag{A8}
\end{aligned}$$

Appendix 4. Proof of Remark 3.1

$$\begin{aligned}
&\text{Corr}(\phi(X_t), R_t | \Omega_t^{\phi, q}) \\
&= \frac{\mathbb{E}_t[\phi(X)R | \Omega_t^{\phi, q}] - \mathbb{E}_t[\phi(X) | \Omega_t^{\phi, q}]\mathbb{E}_t[R | \Omega_t^{\phi, q}]}{\text{Var}(\phi(X_t) | \Omega_t^{\phi, q})^{\frac{1}{2}} \text{Var}(R | \Omega_t^{\phi, q})^{\frac{1}{2}}} \\
&\implies \mathbb{E}_t[\phi(X)R | \Omega_t^{\phi, q}] \\
&= \text{Var}(\phi(X_t) | \Omega_t^{\phi, q})^{\frac{1}{2}} \text{Var}(R | \Omega_t^{\phi, q})^{\frac{1}{2}} \text{Corr}(\phi(X_t), R_t | \Omega_t^{\phi, q}) \\
&\quad + \mathbb{E}_t[\phi(X) | \Omega_t^{\phi, q}]\mathbb{E}_t[R | \Omega_t^{\phi, q}]. \tag{A9}
\end{aligned}$$

Also,

$$\begin{aligned}
\text{Var}(R | \Omega_t^{\phi, q}) &= \text{Var}(\tilde{\mathbf{F}}_t \tilde{\mathbf{Z}}_t + \tilde{\mathcal{E}}_t | \Omega_t^{\phi, q}) \\
&= \tilde{\mathbf{F}}_t^2 \text{Var}(\tilde{\mathbf{Z}}_t | \Omega_t^{\phi, q}) + \text{Var}(\tilde{\mathcal{E}}_t | \Omega_t^{\phi, q}) \\
&= \tilde{\mathbf{F}}_t^2 \text{Var}(\tilde{\mathbf{Z}}_t | \Omega_t^{\phi, q}). \tag{A10}
\end{aligned}$$

As a Result,

$$\begin{aligned}
&\text{Corr}(\phi(X_t), R_t) \\
&= \frac{\mathbb{E}_t[\phi(X)R]}{\text{Var}(R_t)^{\frac{1}{2}}} \\
&= \frac{1}{\text{Var}(R_t)^{\frac{1}{2}}} \sum_{q=1}^Q \mathbb{P}_t(\Omega_t^{\phi, q}) \mathbb{E}_t[\phi(X)R | \Omega_t^{\phi, q}] \\
&= \frac{\tilde{\mathbf{F}}_t}{Q \text{Var}(R_t)^{\frac{1}{2}}} \sum_{q=1}^Q \text{Var}(\phi(X_t) | \Omega_t^{\phi, q})^{\frac{1}{2}} \text{Var}(\tilde{\mathbf{Z}}_t | \Omega_t^{\phi, q})^{\frac{1}{2}} \\
&\text{Corr}(\phi(X_t), R_t | \Omega_t^{\phi, q}) \\
&\quad + \frac{1}{Q \text{Var}(R_t)^{\frac{1}{2}}} \sum_{q=1}^Q \mathbb{E}_t[\phi(X) | \Omega_t^{\phi, q}] \mathbb{E}_t[R | \Omega_t^{\phi, q}]. \tag{A11}
\end{aligned}$$

Appendix 5. Lemma A.1

LEMMA A.1 If $\tilde{\mathbf{z}}\mathbf{v}_e^*$ is the optimal linear predictors under quadratic mean for $\hat{\mathbb{P}}_e$, then $\mathbf{v}_e^* = \mathbb{E}_e[\tilde{\mathbf{F}}]$ for $e \in \{src, trg\}$.

Proof $\mathbf{v}_e^*$ is selected according to the minimization of the following:

$$\begin{aligned}
\mathbb{E}_e[(R - \tilde{\mathbf{Z}}\mathbf{v})^2] &= \mathbb{E}_e[(\tilde{\mathbf{Z}}\tilde{\mathbf{F}} + \tilde{\mathcal{E}} - \tilde{\mathbf{Z}}\mathbf{v})^2] \\
&= \mathbb{E}_e[(\tilde{\mathbf{Z}}(\tilde{\mathbf{F}} - \mathbf{v}))^2] + \mathbb{E}_e[\tilde{\mathcal{E}}^2] - \mathbb{E}_e[\tilde{\mathcal{E}}]\mathbb{E}_e[\tilde{\mathbf{Z}}(\tilde{\mathbf{F}} - \mathbf{v})] \\
&= \mathbb{E}_e[(\tilde{\mathbf{Z}}(\tilde{\mathbf{F}} - \mathbf{v}))^2] + \mathbb{E}_e[\tilde{\mathcal{E}}^2]. \tag{A12}
\end{aligned}$$

Then, we would like to find $\mathbf{v}_e^*$ that yields $D_{\mathbf{v}}\mathbb{E}_e[(R - \tilde{\mathbf{Z}}\mathbf{v})^2] = 0$.

$$\begin{aligned}
D_{\mathbf{v}}\mathbb{E}_e[(R - \tilde{\mathbf{Z}}\mathbf{v})^2] &= D_{\mathbf{v}}\mathbb{E}_e[(\tilde{\mathbf{Z}}(\tilde{\mathbf{F}} - \mathbf{v}))^2] \\
&= -2\mathbb{E}_e[\tilde{\mathbf{Z}}^2(\tilde{\mathbf{F}} - \mathbf{v})] \\
&= -2\mathbb{E}_e[\mathbb{E}_t[\tilde{\mathbf{Z}}^2]](\mathbb{E}_e[\tilde{\mathbf{F}}] - \mathbf{v}) \\
&= -2(\mathbb{E}_e[\tilde{\mathbf{F}}] - \mathbf{v}). \tag{A13}
\end{aligned}$$

Consequently, (A13) vanishes if we choose $\mathbf{v}_e^* = \mathbb{E}_e[\tilde{\mathbf{F}}]$. ■

Appendix 6. Hyperparameters & model diversity

Table A1. Hyperparameters for ranking-aware MSE.

	LSTM	ResNet1D	TFM	HIST	CoarHGNN	GAT
β_{base}	0.2	0.1	0.2	0.5	0.5	0.3
lr	$2e-6$	$2e-5$	$2e-5$	$2e-6$	$1e-5$	$2e-6$

Table A2. Hyperparameters for the Causal Discovery Objective.

	LSTM	ResNet1D	TFM	HIST	CoarHGNN	GAT
β_1	10	10	2	10	10	10
β_2	2	2	0.5	5	5	2
lr	$5e-5$	$2e-6$	$5e-5$	$2e-5$	$2e-5$	$5e-5$

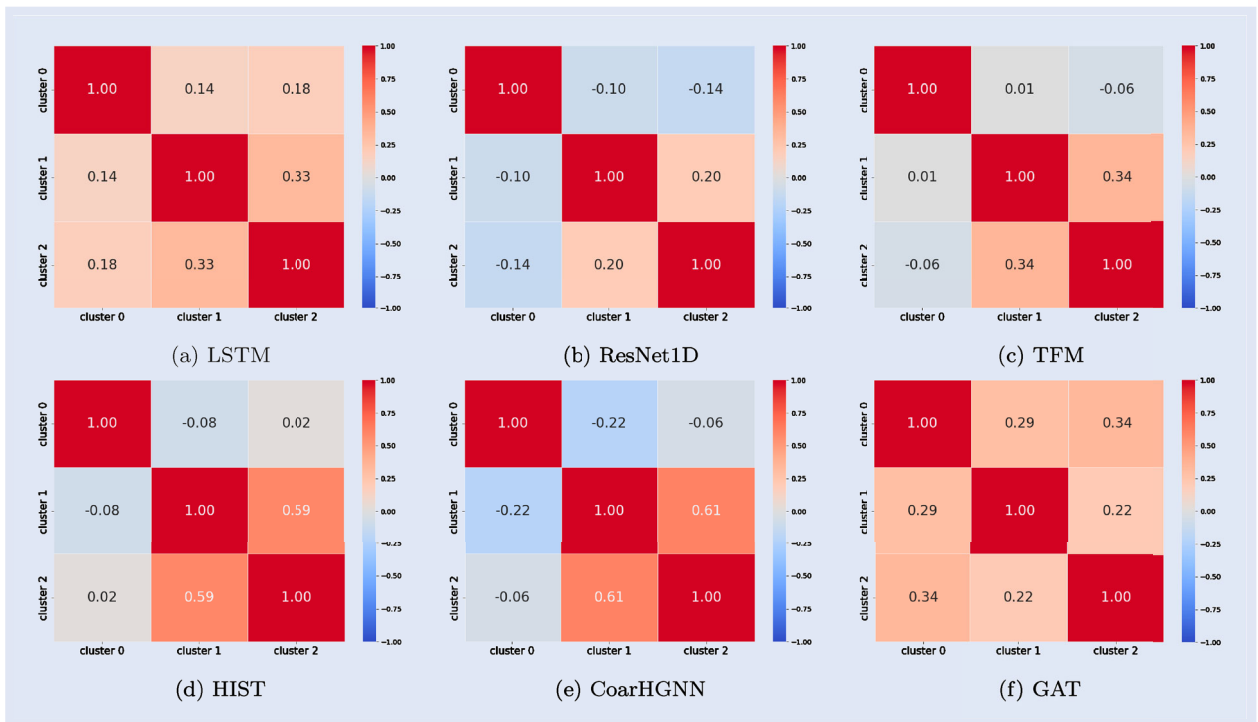


Figure A1. The correlations of cluster centers in figure 10. (a) LSTM. (b) ResNet1D. (c) TFM. (d) HIST. (e) CoarHGNN and (f) GAT.